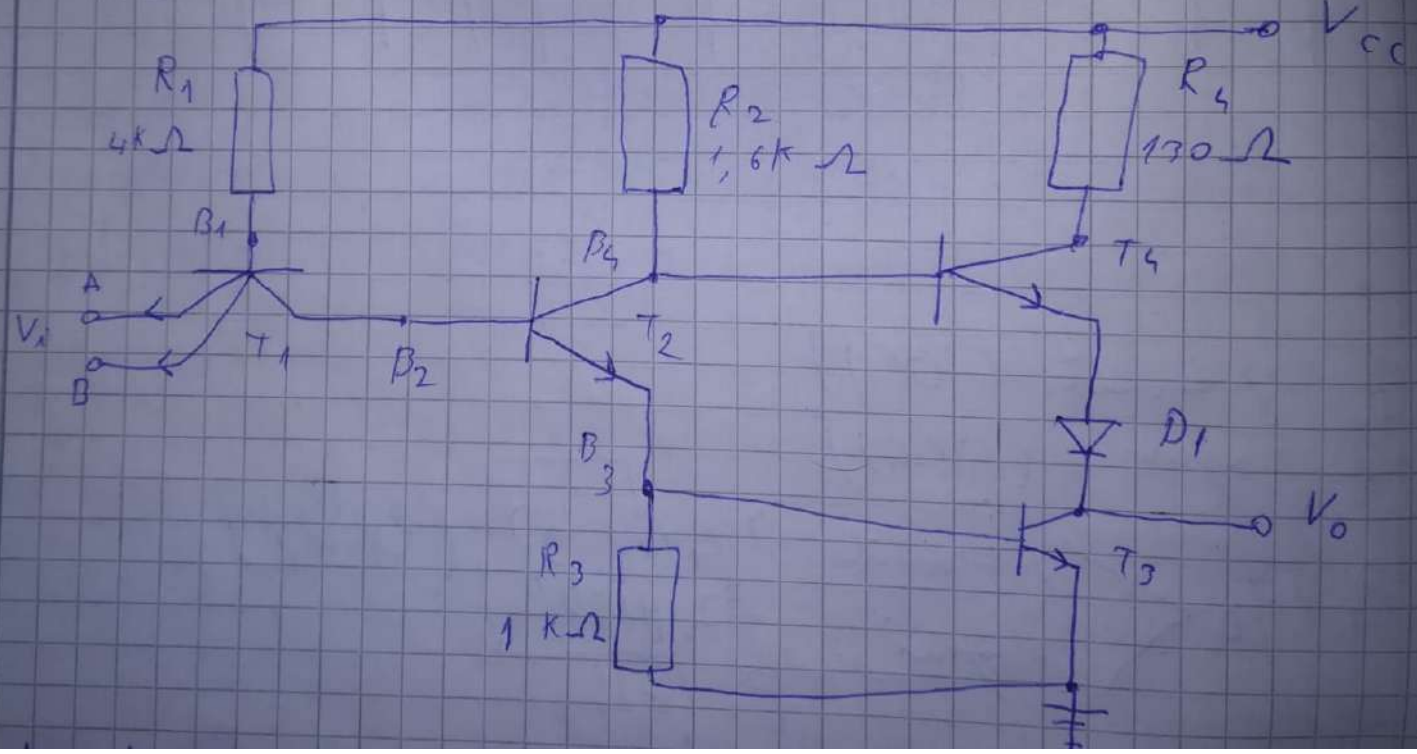
Aplicația 1 CDI - complet:

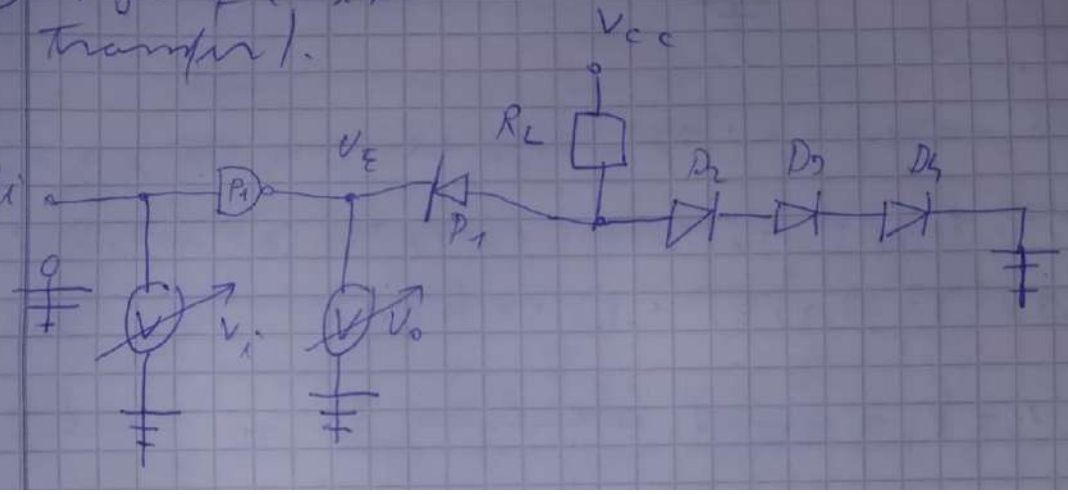
1. Scop: metodă de determinare a parametrilor statici la circuitele integrate TTL.

2. Teorie:



3. Mod de lucru:

(1) $V_0 = f(V_i)$. (ridicarea caracteristicii statice de transfer).



$V_o = f(V_i)$ pt. $V_{cc}$ const., $R_L$ variabil

$V_o = f(V_i)$ pt. $R_L$ const., $V_{cc}$ variabil.

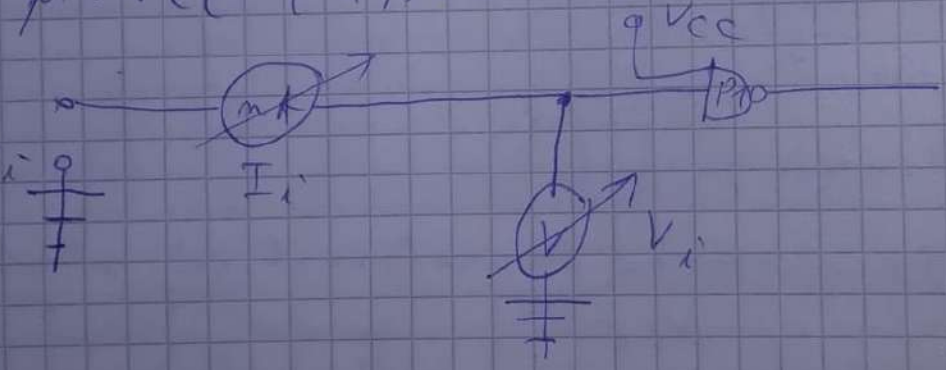
$V_I[V] = \{0; 0,1; 0,2; 0,3; 0,4; 0,5; 0,6; 0,7; 0,8; 0,9; 1; 1,2; 1,3; 1,4; 1,5; 1,6; 1,7; 2; 2,75; 3; 3,5; 4; 4,5; 5; V_{cc}\}$

a) pt. $R_L = 400\,\Omega$ ; $V_{cc} = \{4,5\,V; 5\,V; 5,5\,V\}$.

b) pt. $V_{cc} = 5\,V$; $R_L = \{400\,\Omega; 800\,\Omega; 4\,K\Omega\}$

(II) $I_i = f(V_i)$ (caracteristica de intrare)

pt. $V_{cc} = \{4,5\,V; 5\,V; 5,5\,V\}$.




$V_I[V] = \{0; 0,1; 0,2; 0,3; 0,4; 0,5; 0,6; 0,7; 0,8; 0,9; 1; 1,1; 1,2; 1,3; 1,4; 1,5; 1,6; 1,7; 1,8; 1,9; 2; 2,2; 2,3; 3; 3,5; 4; 4,5; 5; V_{cc}\}$.

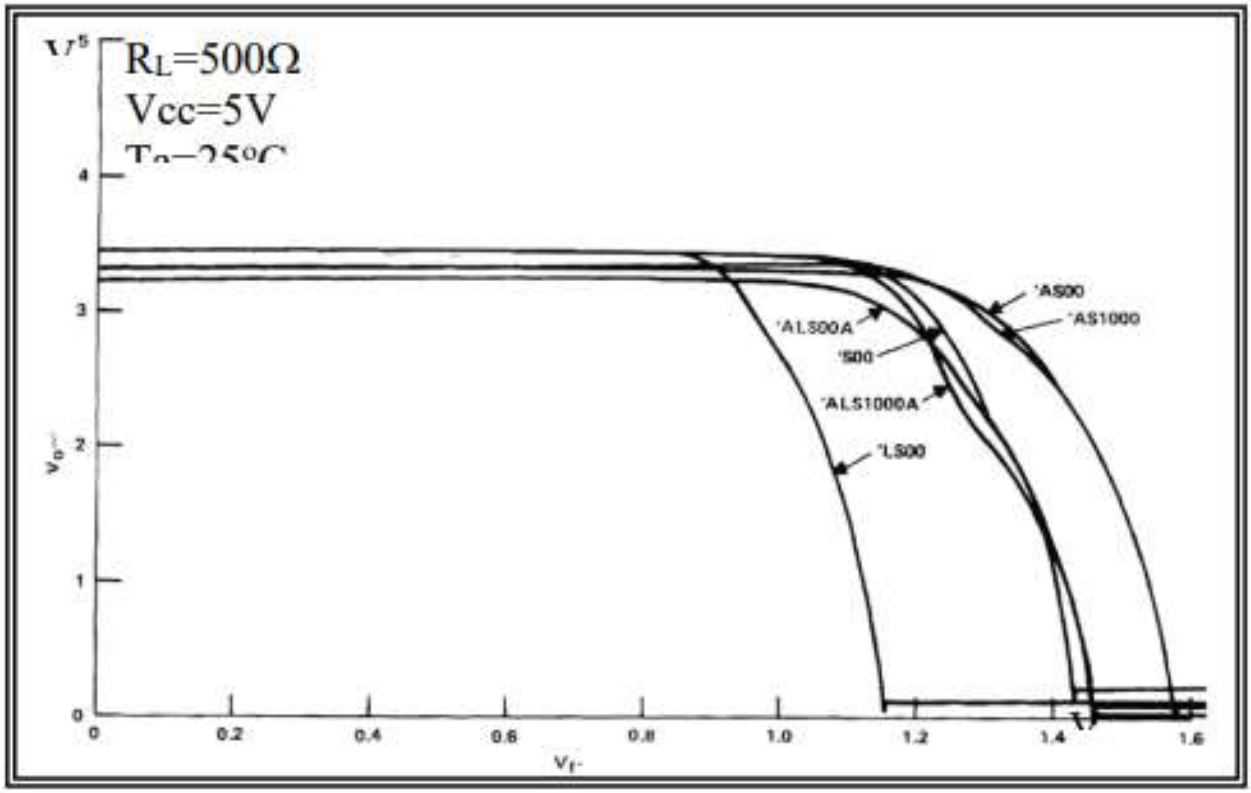



Fig. A1.4
Caracteristica statică de transfer pentru unele circuite integrate TTL

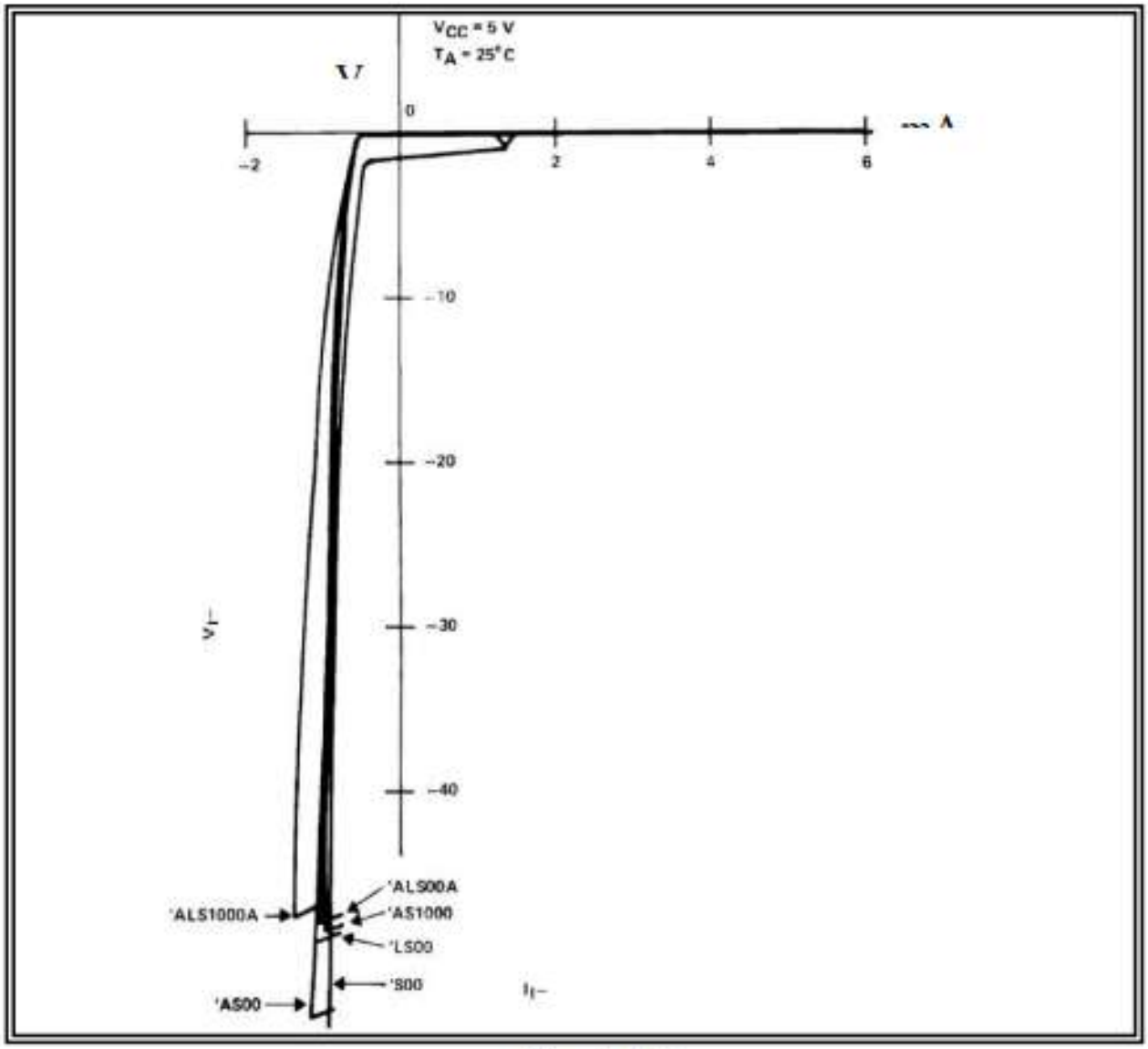



Fig. A1.5
Caracteristica statică de intrare la circuitele integrate TTL

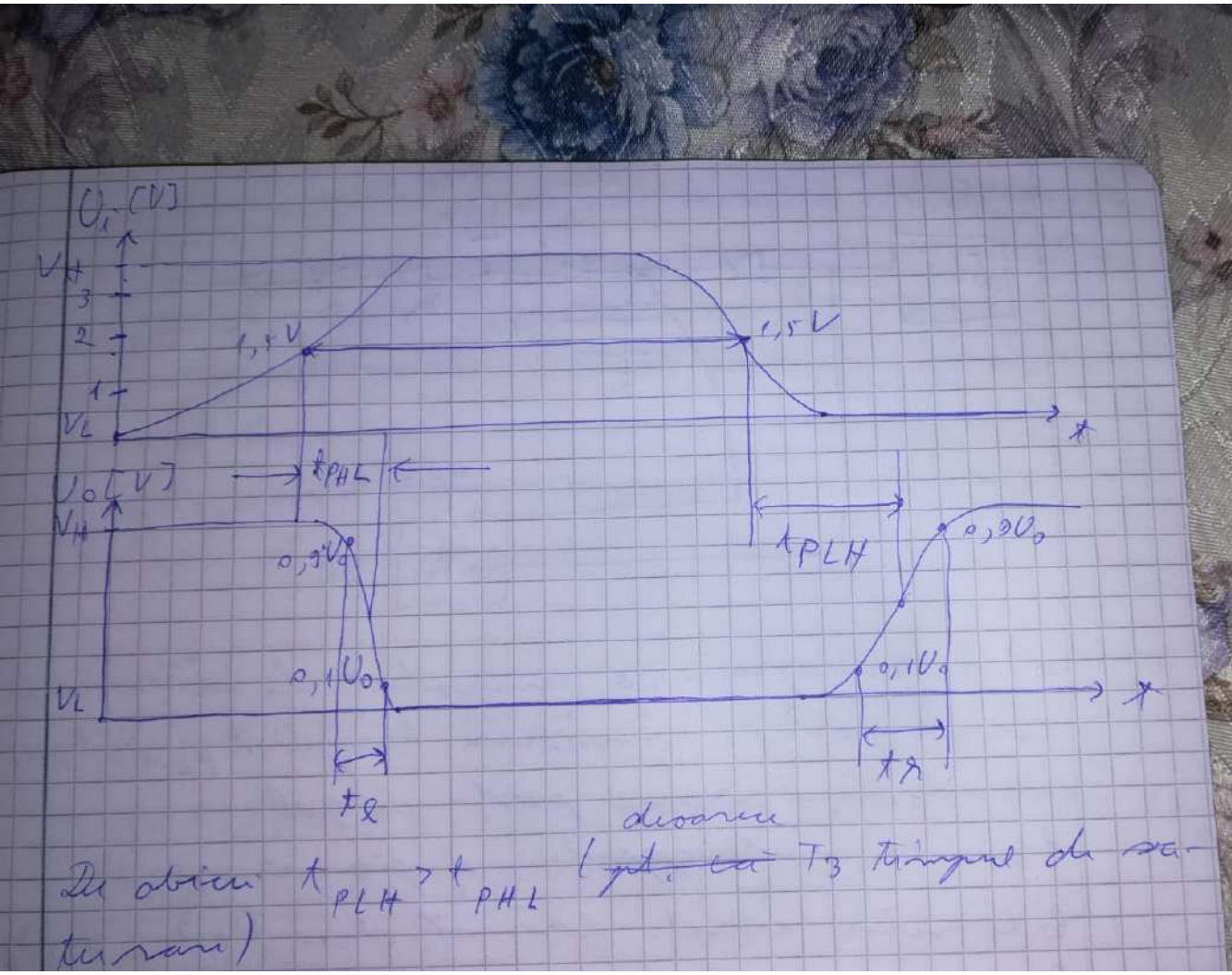
Aplicația 1 CI - compus:

1. Scop: măsurarea timpilor de propagare cât și influența pe care îl produce diferită fanto. asupra acestora.

2. Teorie:

timp de propagare: viteza de comutare a circ. logice.

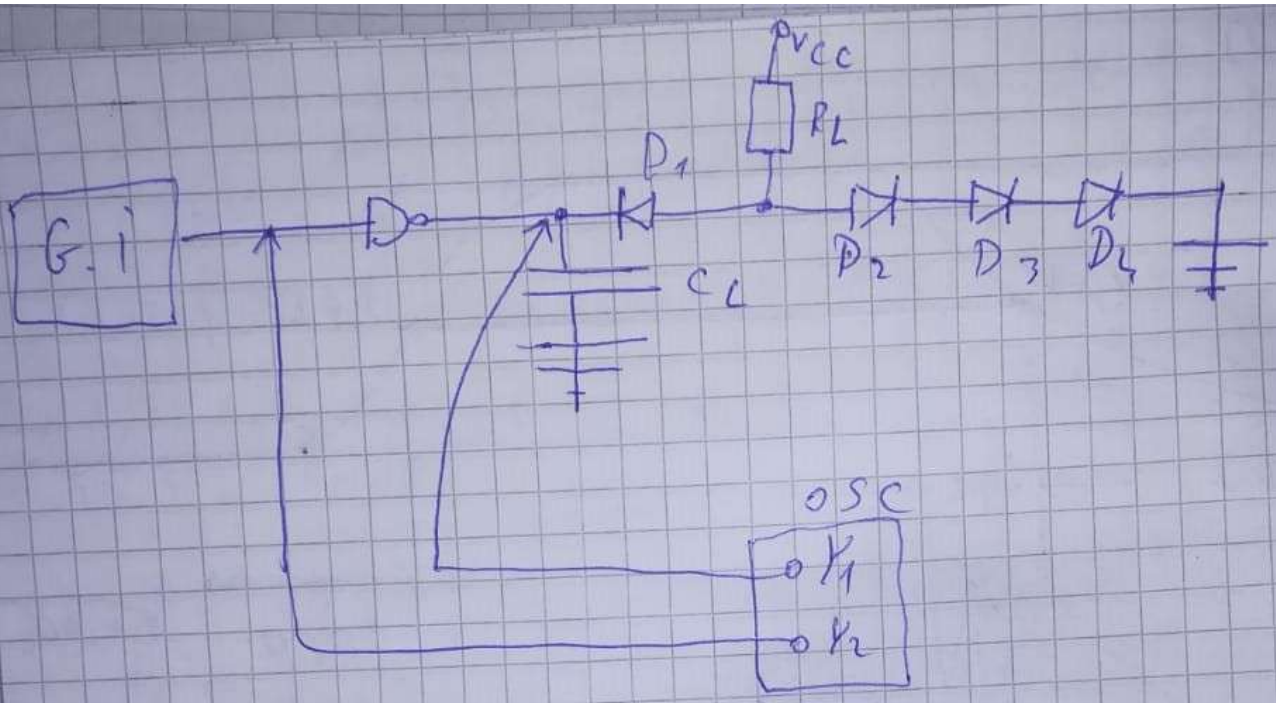
$t_{pd} = \frac{t_{PHL} + t_{PLH}}{2}$ timp mediu de propagare




Deci avem $t_{PLH} > t_{PHL}$ (~~pt. ca~~ durata $T_3$ timpul de saturare)

3. Mod de lucru:

$t_i = 500$ ms; amplitudinea 3,5 V, semnal cu durata de repetiție 1 μs.




a) se ortografiază semnalul de intrare și ieșirea porții; măsurăm timpii $t_{PLH}$, $t_{PHL}$ considerând ieșirea porții în gol.

! canalul osciloscopului să fie sincron.

b) comutare circ. format din $D_1, D_2, D_3, D_4$, $R_L$.

$R_L = \{400\Omega, 800\Omega, 4K\Omega\}$

$V_{CC} = 5V$.

măsurăm timpii:

$t_{PHL} = f(R_L)$

$t_{PLH} = f(R_L)$

$t_{PD} = f(R_L)$

$D_1$ → jonctiune bază-emitor $T_1$.

$D_2, D_3, D_4$ → jonctiune bază-colector $T_2$, și joncț. emitor-bază $T_2$, respectiv $T_3$.

c) conectam circ. de sarcina $D_1, D_2, D_3, D_4, R_L =$
$= 400\Omega$ + simultan capacitati parazitare
$C_L = 4 \times 10 \times pF; 680 pF; 1,5 \times nF; 10 \times nF$ +
osciloscopul semnale de intrare și ieșire
a portii.
- cu ajutorul osciloscopului se măsoară
timpii de propagare.

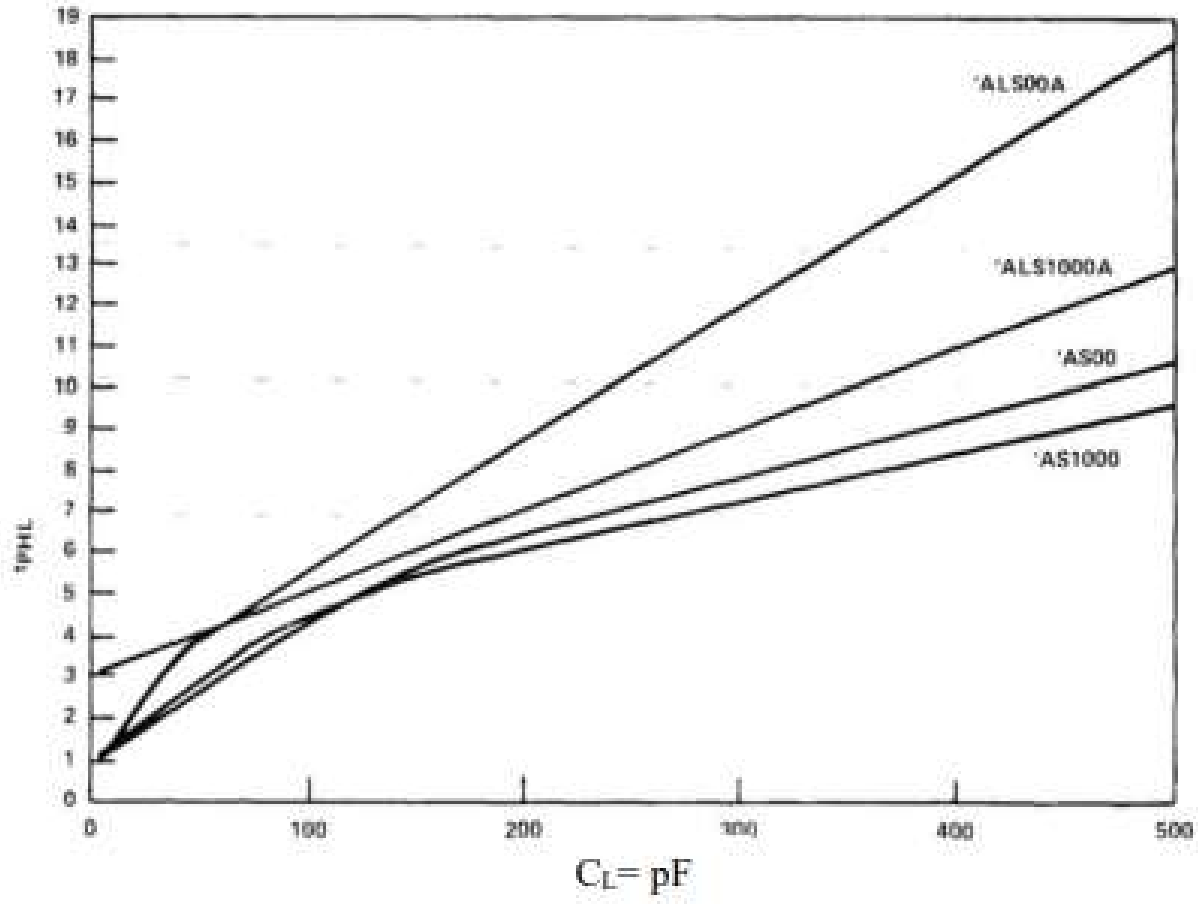



Fig. A2.3
Timpul de propagare $t_{PHL}$ funcţie de capacitatea de sarcină

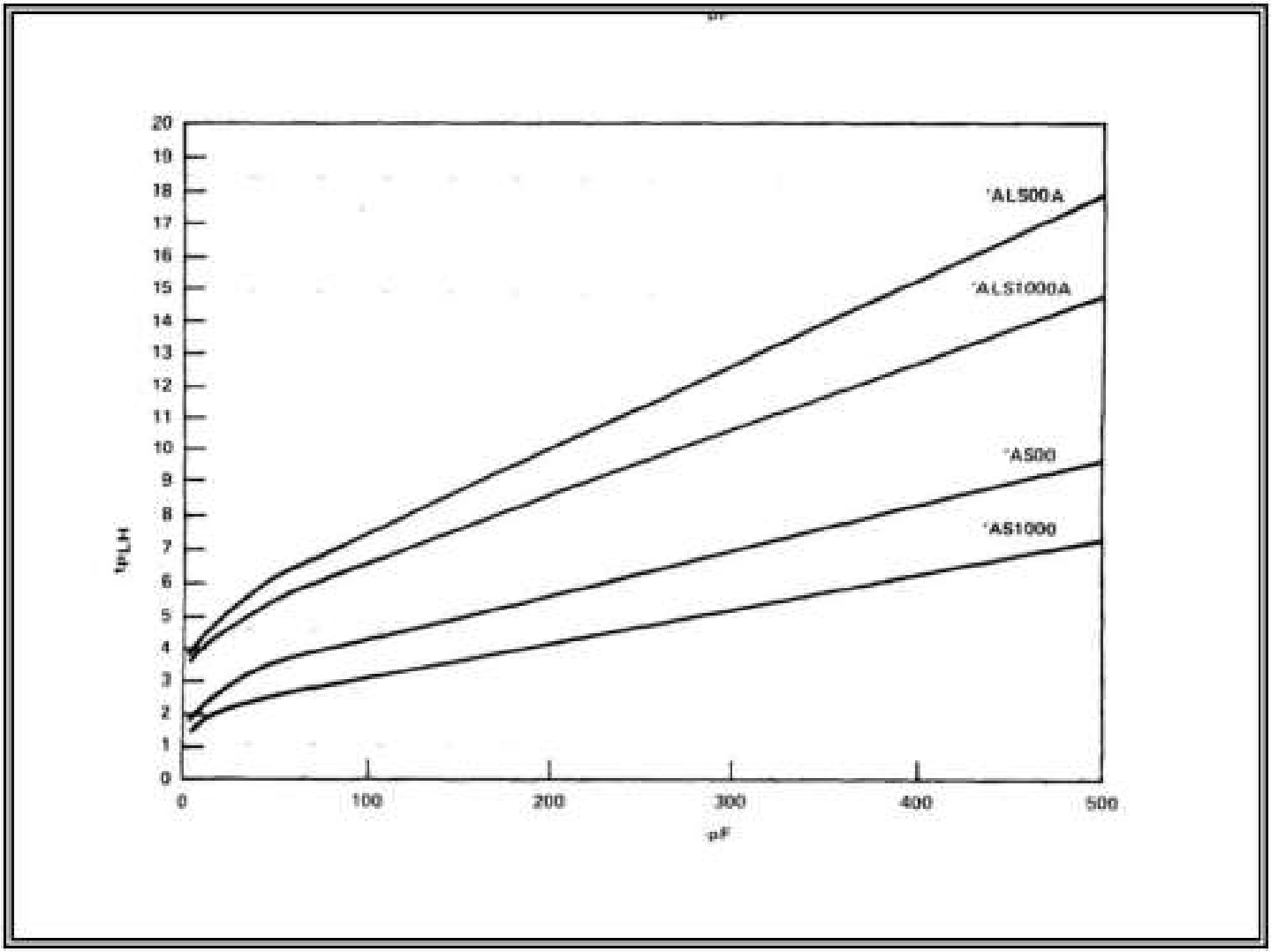



Fig. A2.4
Timpul de propagare $t_{PLH}$ funcție de capacitatea de sarcină

Aplicația nr. 3 - CI; - compact

Circuite basculante astabile cu porți TTL

1. Scop:

- studiul circuitelor basculante astabile realizate cu o poartă TTL, 2 porți + analiza

principii de bază a funcționării.

2. Teorie:

- două inversoare / două porți cu inversare
~~două porți~~ ( $x = \overline{x}$, sau $= \overline{x}$ ).

$\tau = RC$

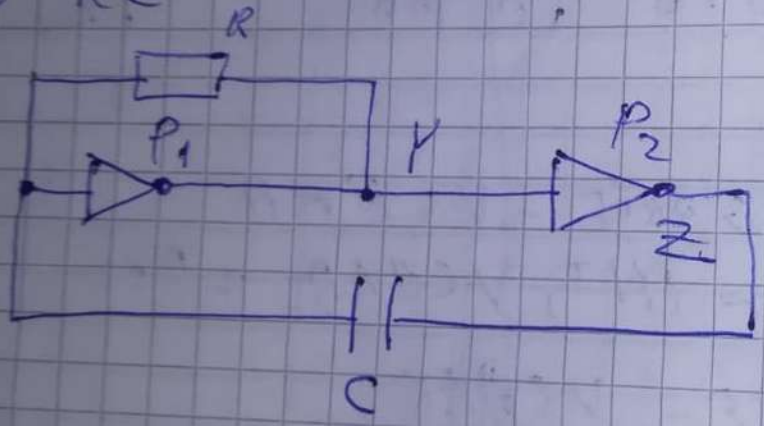
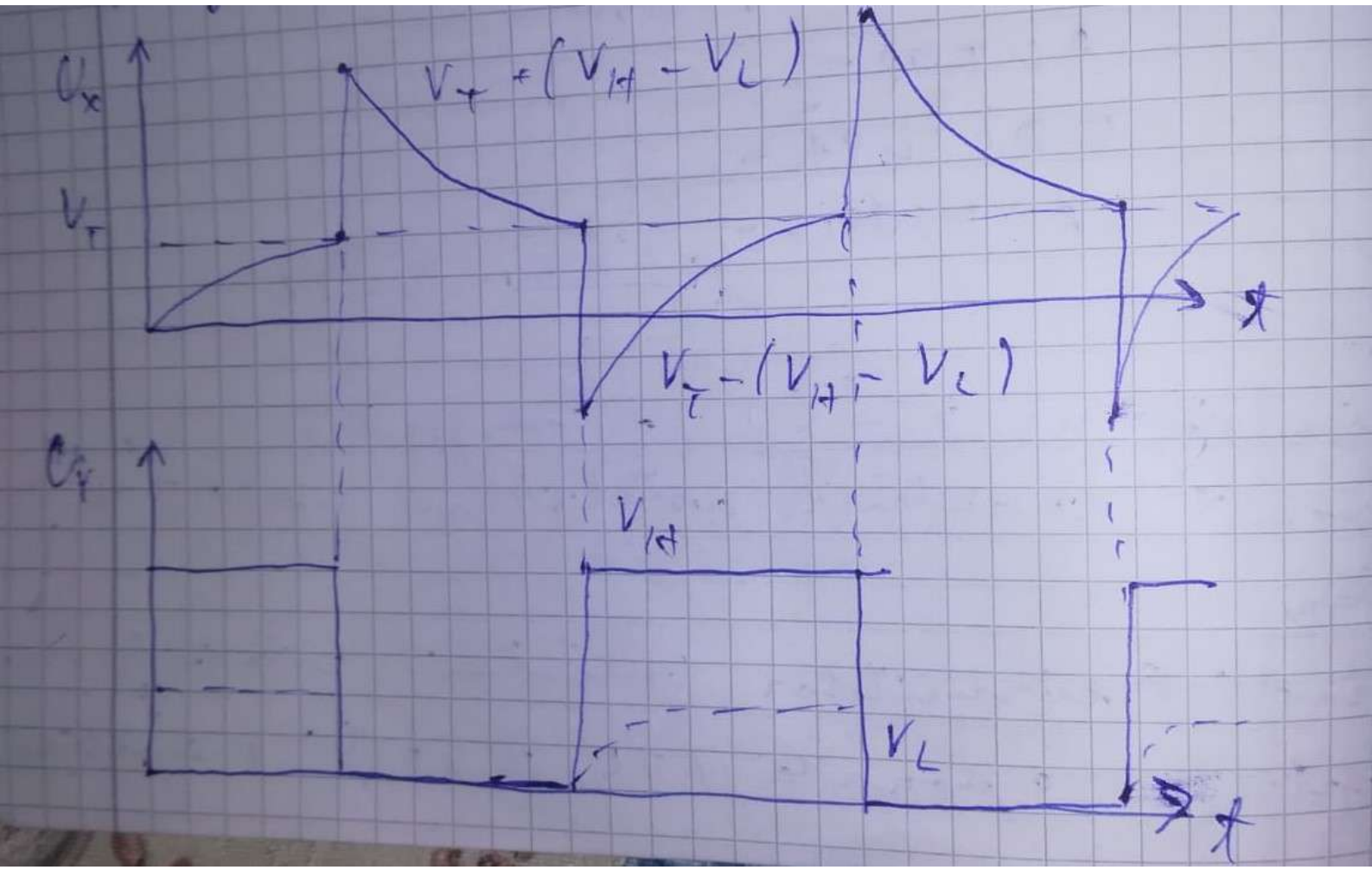
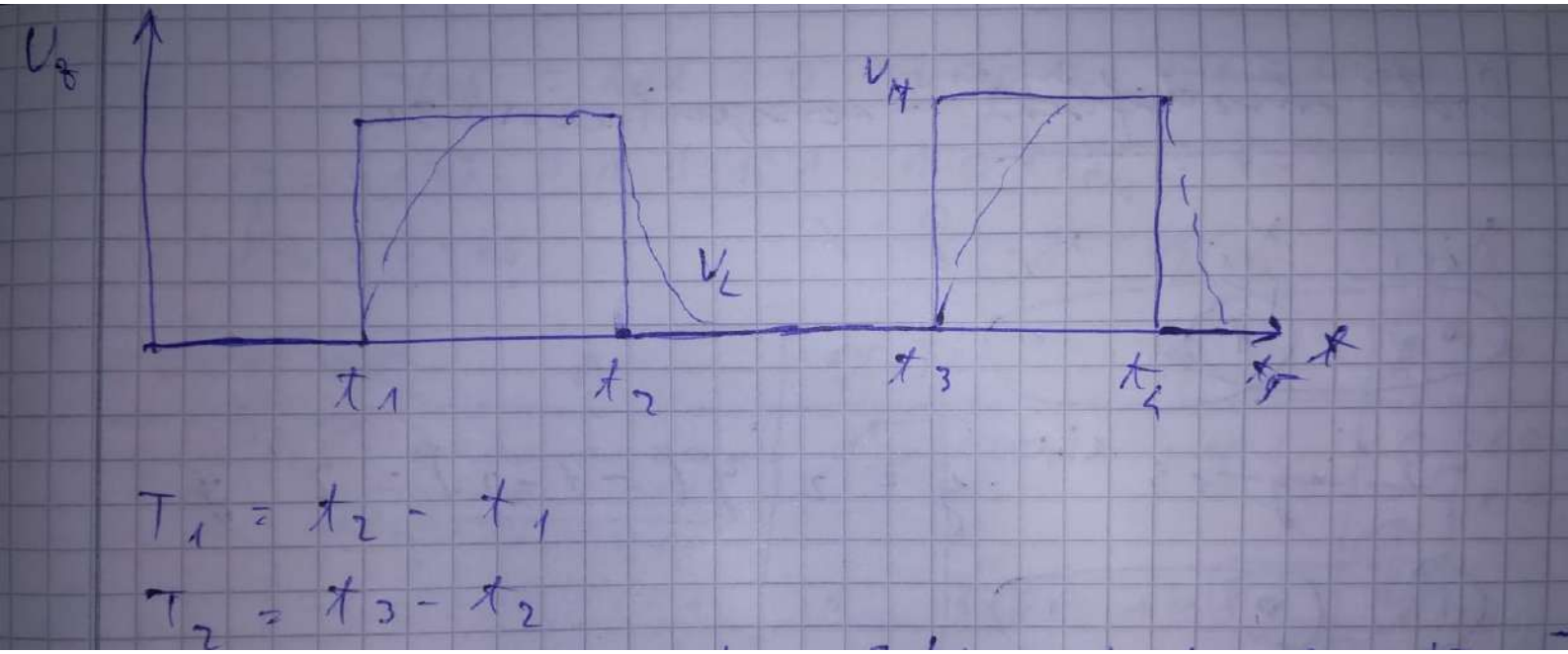



fig 1


$C_x$
$V_T$
$V_T + (V_{IH} - V_L)$
$V_T - (V_{IH} - V_L)$
$x$
$C_y$
$V_{IH}$
$V_L$
$x$




$T_1 = t_2 - t_1$

$T_2 = t_3 - t_2$

$$T_1 = RC \cdot \ln \frac{V_T + V_H - 2V_L}{V_T - V_L}$$

$$T_2 = RC \ln \frac{V_T + V_2 - 2V_{th}}{V_T - V_{th}}$$

3. Mod de lucru:

a) realizam montaj din fig. 1

$R = 390\ \Omega$, $C = 10\ nF$

- osilograma X, Y, Z (memorizam pt Y)

b) $V_{CC} = 5\ V$, $R = \{100, 200, 300, 390\}\ \Omega$

$C = 10\ nF$ - măsurăm duratele $T_1$, $T_2$ și pt. $T$; $T = f(R)$, $C$ - const. (verificăm datele teoretice).

c) $V_{CC} = 5\ V$, $C = \{100\ pF, 500\ pF; 1,5\ nF; 5\ nF; 10\ nF\}$, $R = 390\ \Omega$ - măsurăm duratele $T_1, T_2$ și $T$; $T = T_1 + T_2 = f(C)$, $R$ - const. (verificăm datele experimentale).

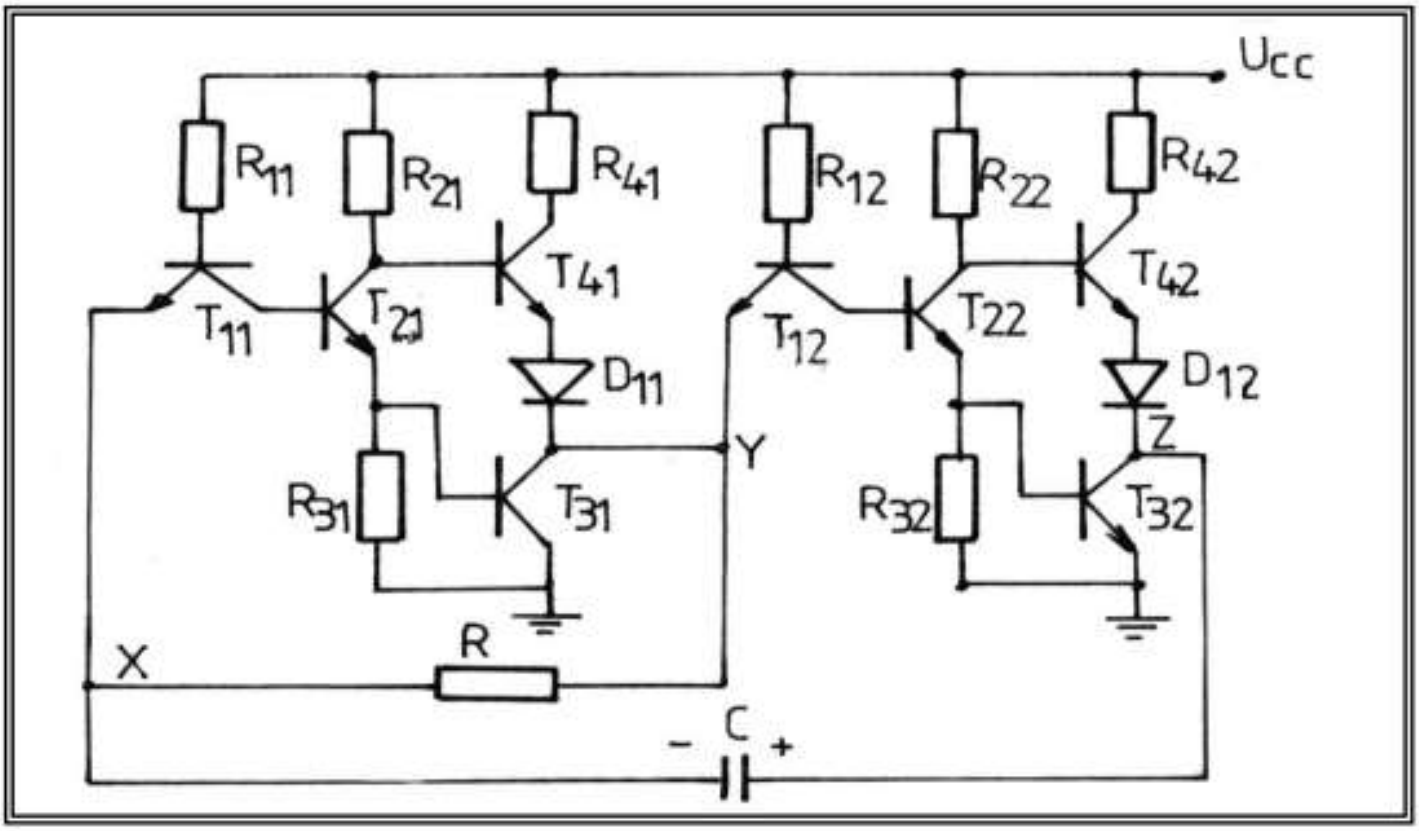
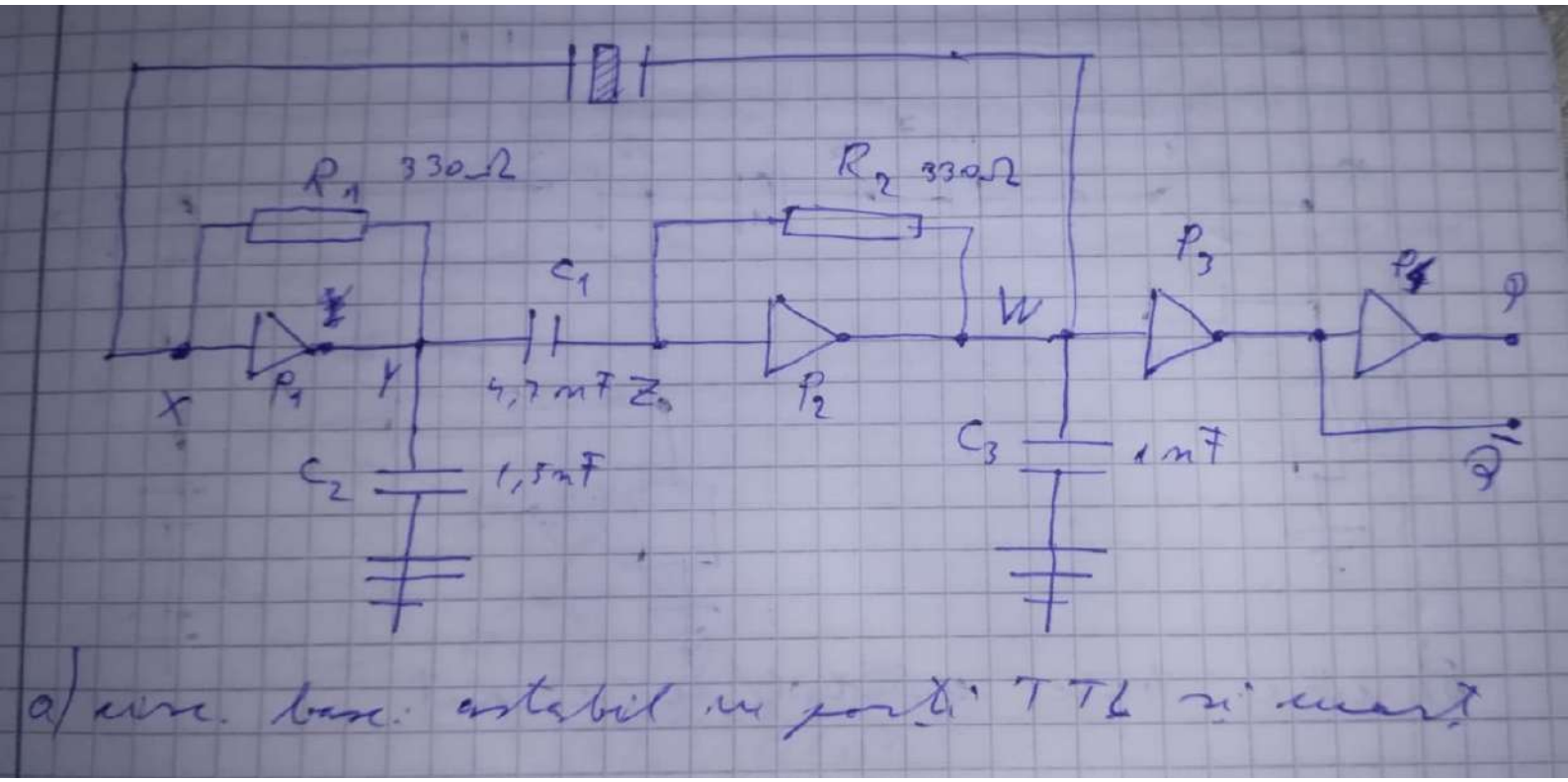


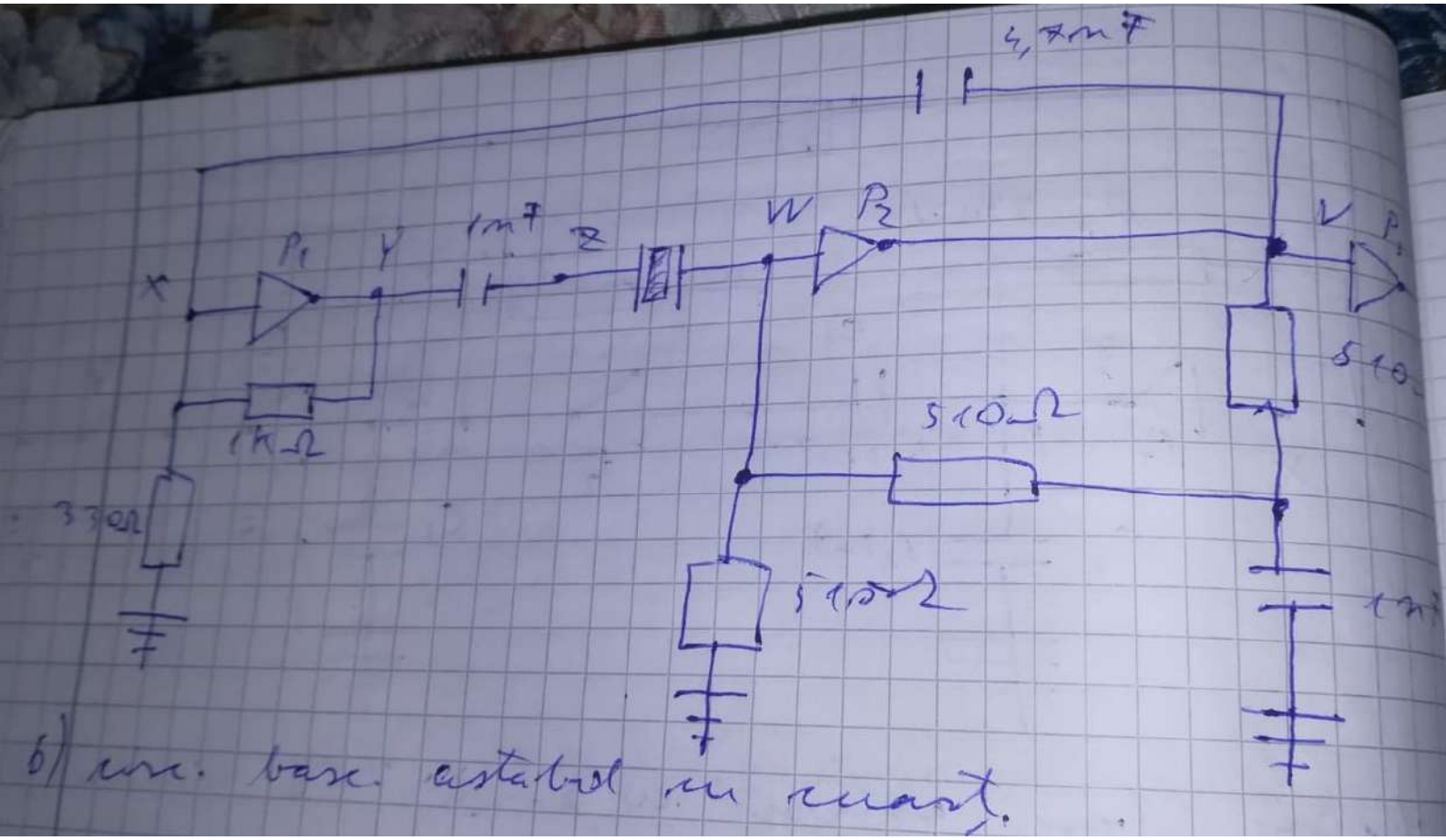

**Fig. 2 Schema electronică internă a circuitului basculant bistabil**

Lab 4 (conspect):

- Scop: studiul circuitelor basculante astabile realizate cu porți TTL și cu cristal de cuarț




a) circ. baz. astabil cu porți TTL și cuarț




6) conf. base. astabil cu cuarț.

3. Mod de lucru:

a) - osciloscopul semnale pe $x, y, z, w, g, q$ (sincronizare pe $g$), pe rând conținut:
- $P_1, P_2, P_3, P_4$ inversate

b) - osciloscopul semnale pe $x, y, z, v, g$
- conținut → siguranță de funcționare
- $P_1, P_2, P_3$ inversate.

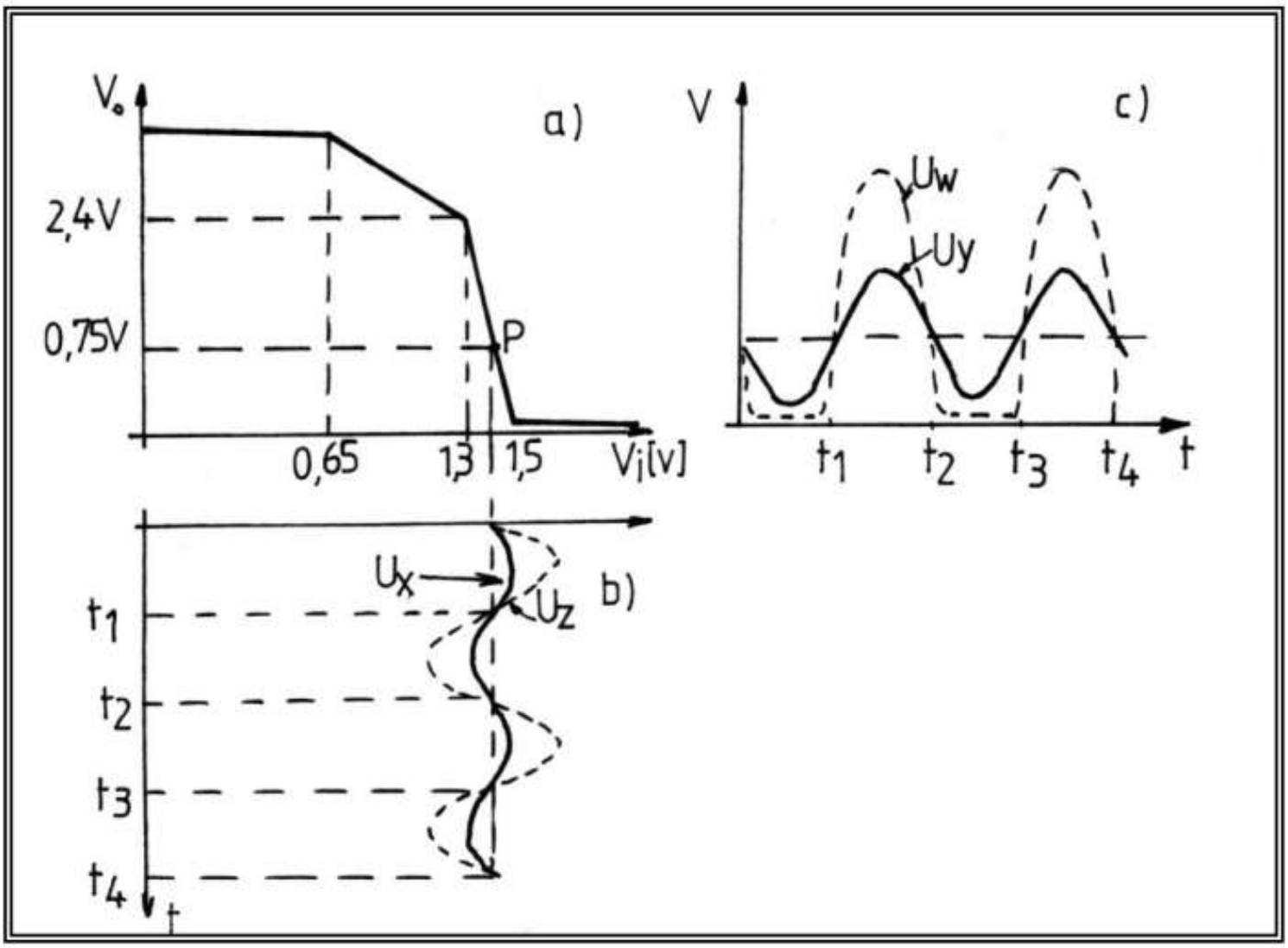



Fig. A7.2

Funcționarea CBA cu cuarț

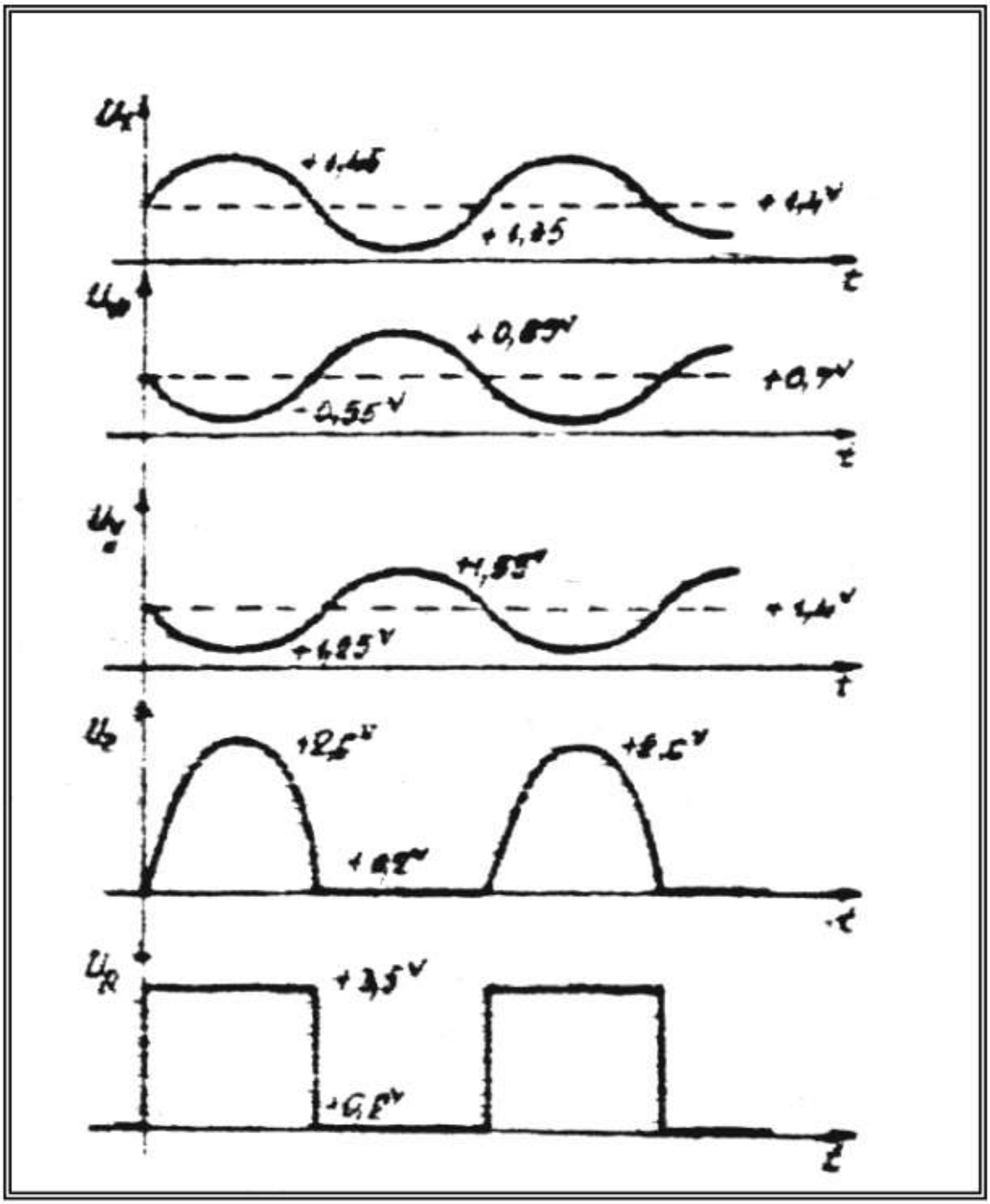


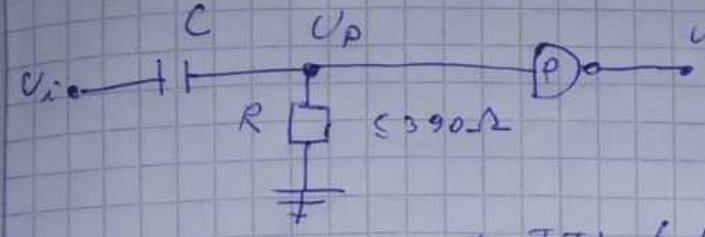
Fig. A7.4

Diagramele de timp de funcţionare a CBA cu cuarț și porți TTL

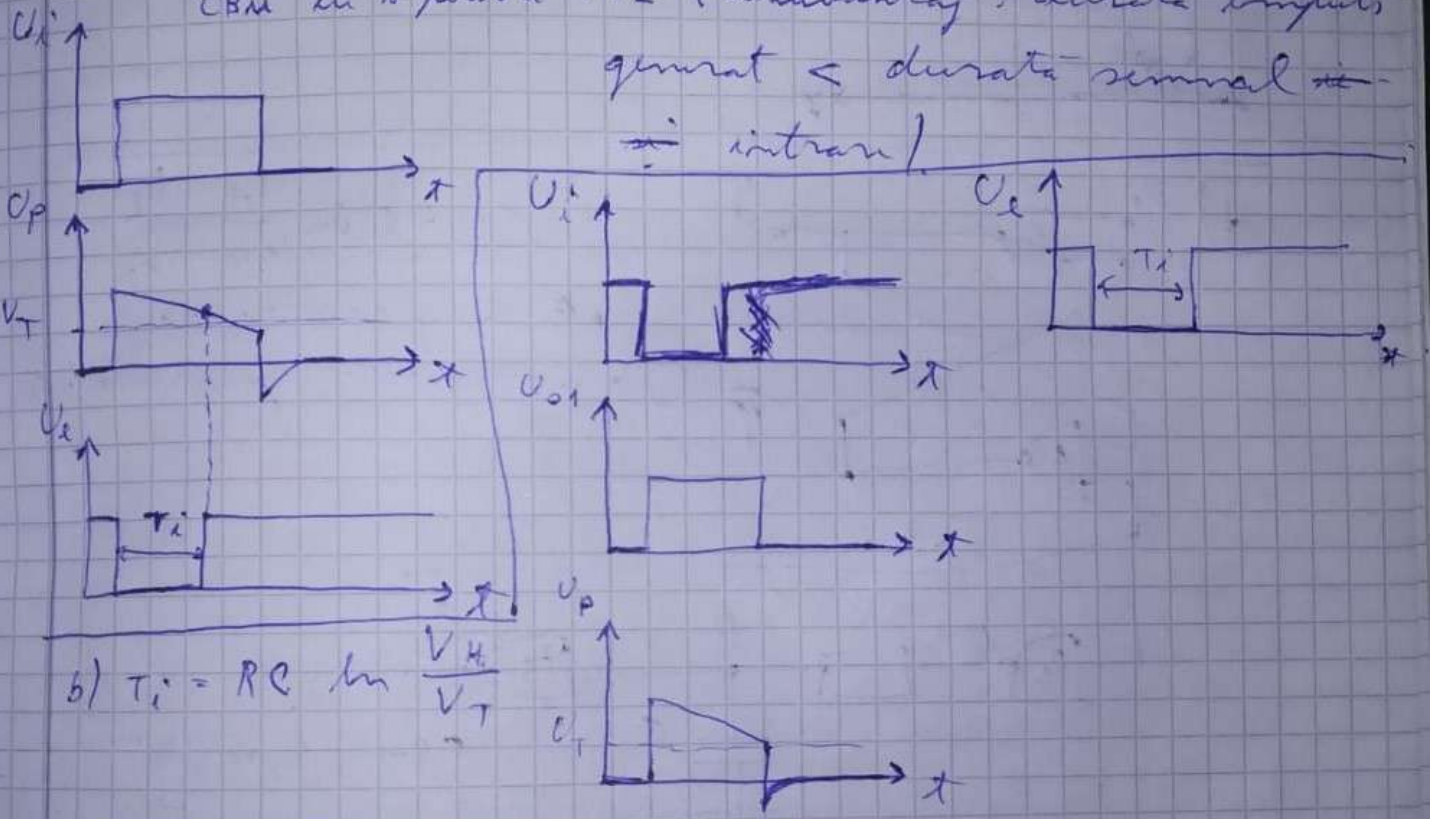
Aplicația 5 CI (continuă):

1. Scop: circ. bază monostab. realizați prin porți TTL; front ridicător și căzător al semnalului de comandă.

2. Teorie: a) $T_i = RC \ln \frac{V_{IH} - V_L}{V_T}$

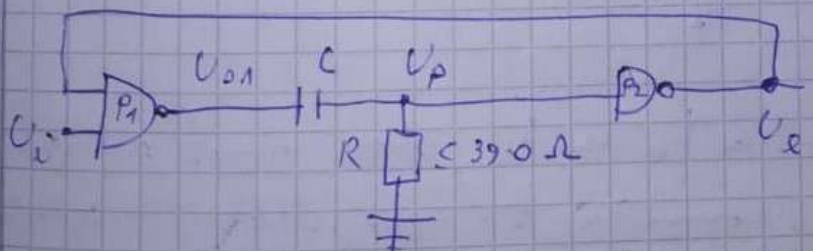



CBM cu o poartă TTL (dezavantaj: durata impulsului generat ≤ durata semnalului intrare)



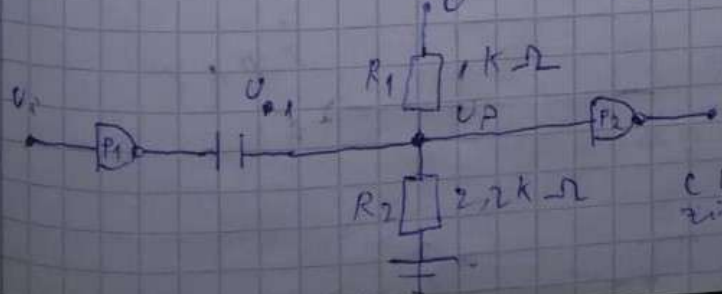

b) $T_i = RC \ln \frac{V_H}{V_T}$



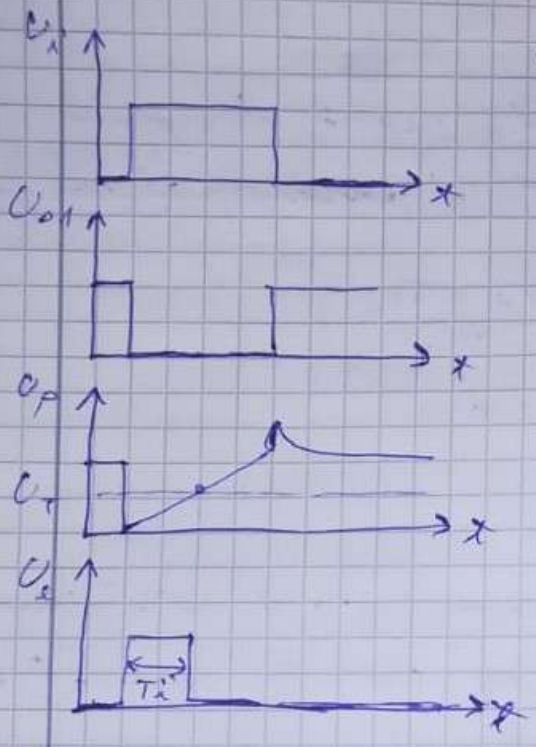




CBM cu două porți, front negativ al semnalului de comandă

c) $T_i = R_1 C \ln \frac{V_p - (V_H - V_L) - V_{cc}}{V_T - V_{cc}}$

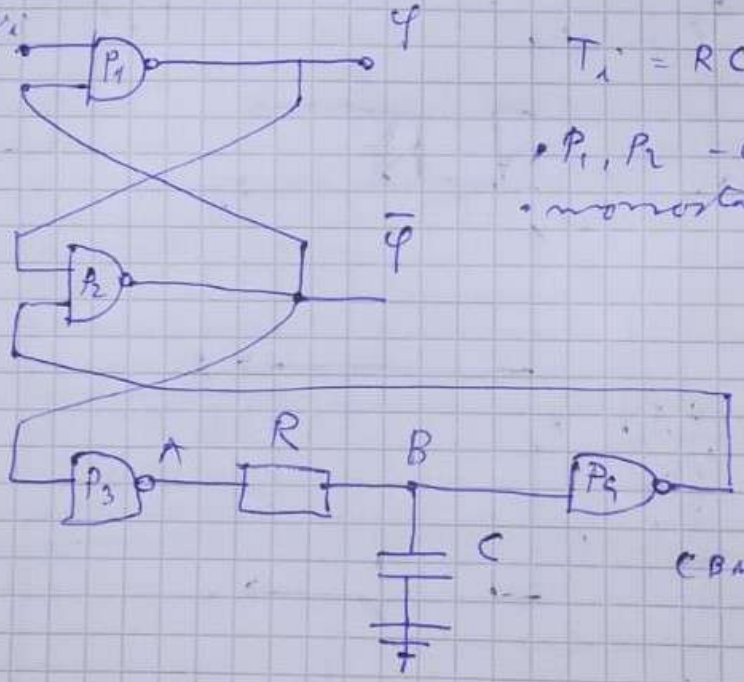



CBM cu două porți; front po-
zitiv —"—




• în regim stabil tensiunea din tranzistorul ce nu este în saturație este mai mare decât tensiunea de prag a $P_2$

d)




$$T_i = RC \ln \frac{V_H - V_L}{V_H - V_T}$$

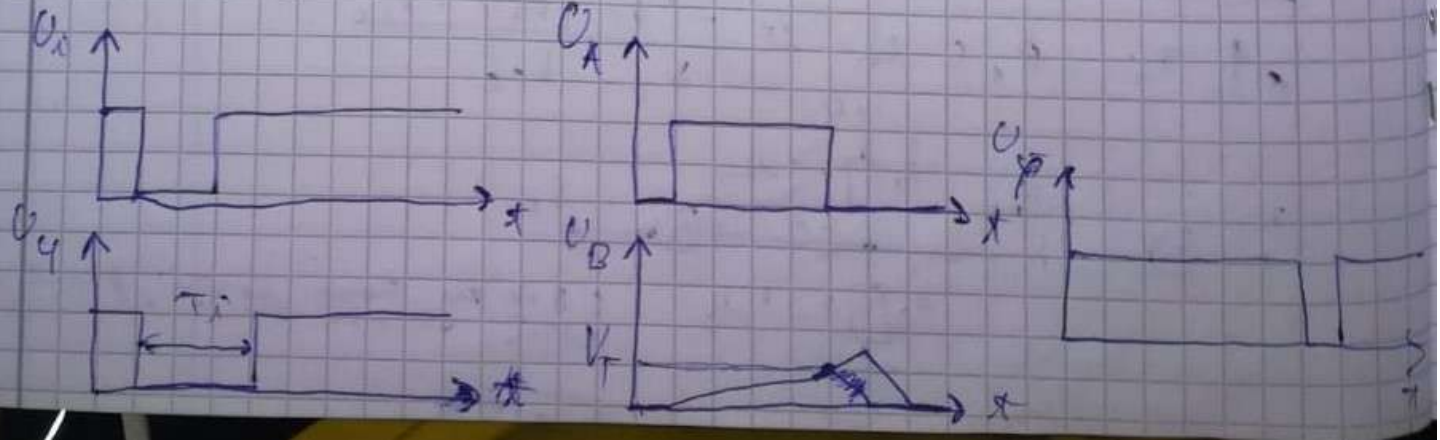
• $P_1, P_2$ - CBD și RS

• monostabil propriu-zis

CBM - 4 porți

• C se încarcă de la $V_L$ la $V_H$, dar când devine egală cu $V_T$ a $P_4$, comută => comută basculul RS.

• după comutare, C se descarcă rapid prin ieșirea din ieșirea al $P_3$.



3. Mod de lucru:

a) • montaj; diagrama în P și ieșire.
• în fct. de val. $T_i$ și alți parametri de timp ai semnalului de intrare (determinați teoretic a RC).
• durată semnal de intrare va fi mai mare decât $T_i$; analog semnalul de ieșire.

b) • montaj; diagramă; durata $T_i$; comparare rezultate ~~din~~ etapa ~~aspect~~ teoretică cu cel experimental.

c) C mare (ex: 100 nF) ⇒ constantă RC mare; diagrama (accent pe tranziția 0→1); observații; comparare cu c).

d) C = 10 nF; montaj; diagrama; $T_i$.

e) R = 390 Ω, C = 10 nF; montaj; diagrama; $T_i$; comparare cu rezultate teoretice
R = 390 Ω, C = 100 nF; diagrama; comparare cu cele de la c); observații

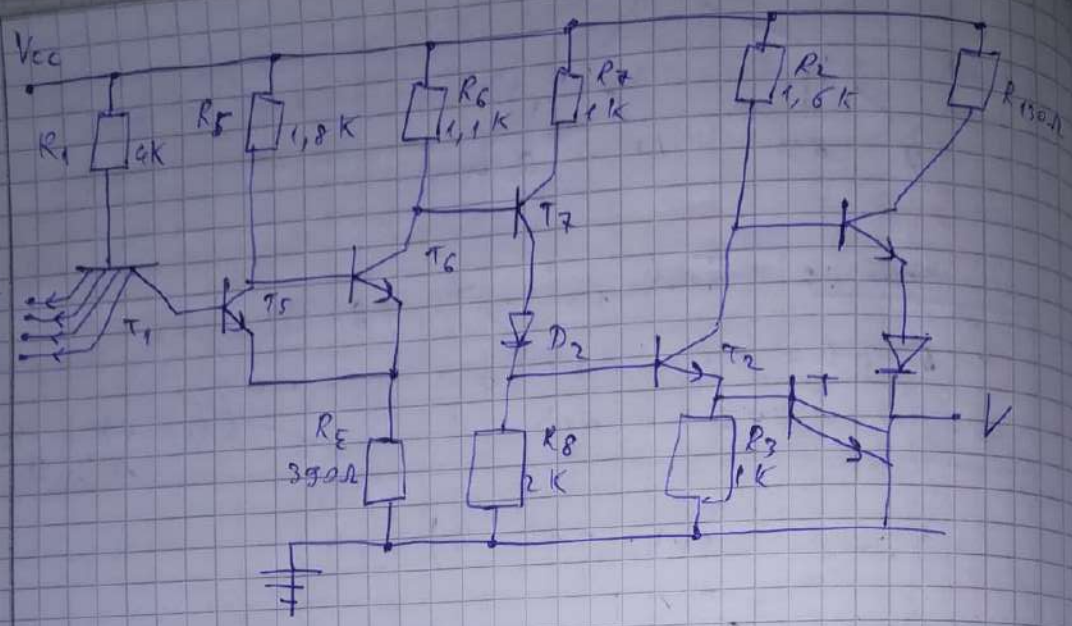
x = 2398 = 0000 1001 0101 1110
y = 19 = 0000 1001 1

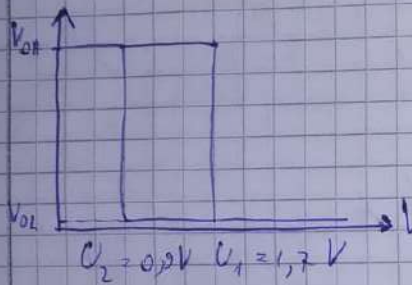
Aplicația 6 CI (complet):

1. ~~Scop~~ Scop: studiul triggerelor Schmitt integrați TTL d.p.d.v. al comportării în regim dinamic cât și unele aplicații practice.

2. Teoria lucrării:

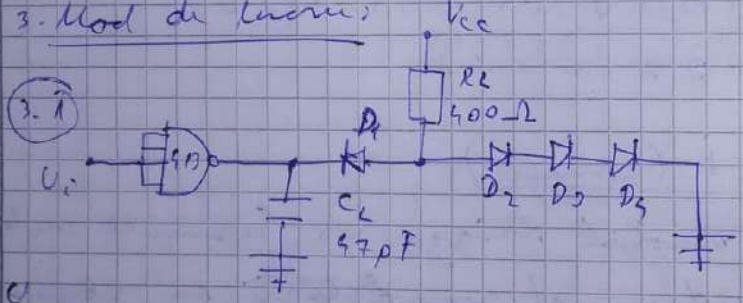
- pe lângă funcția sa de formator, circ. îndeplinește fct. logică ȘI-NU; A, B, C, D - var. de intrare
- circ. are pentru fiecare intrare același model ca la o poartă TTL.
- trigger → $T_5$, $T_6$.
- colector $T_5$ și baza $T_6$ (cuplaj se face direct).

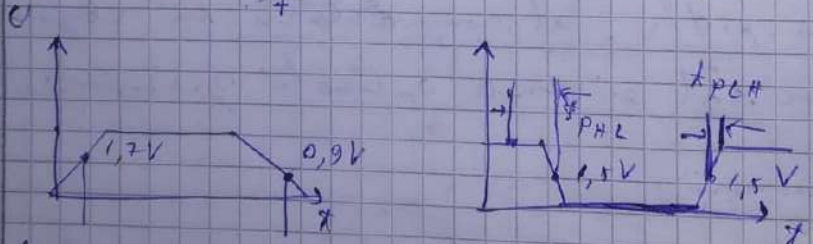



3. Mod de lucru:

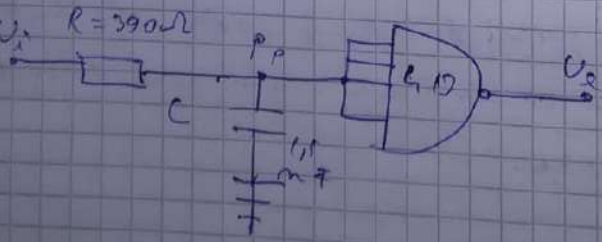
3.1



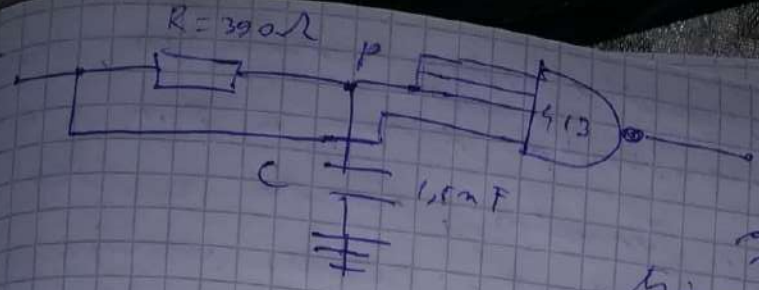




↳ mașuran trigger Schmitt.

3.2




aplicație ✱

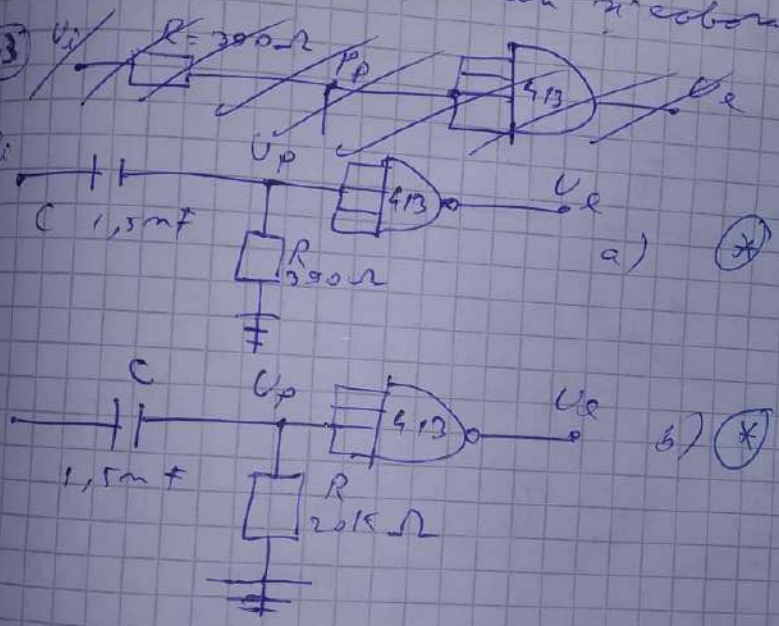



aplicația 2 (*)

Trigger Schmitt

- realizare montaj + osciloscopul semnal de intrare și ieșire.
- măsurare timpi $t_{pAL}$, $t_{pLH}$.
- $t = 3,5V$; durată apăsării 1P; durata impuls 500 ms; front ridicare coborâre $t_R = t_c = 100ms$.

3.3




a) (*)

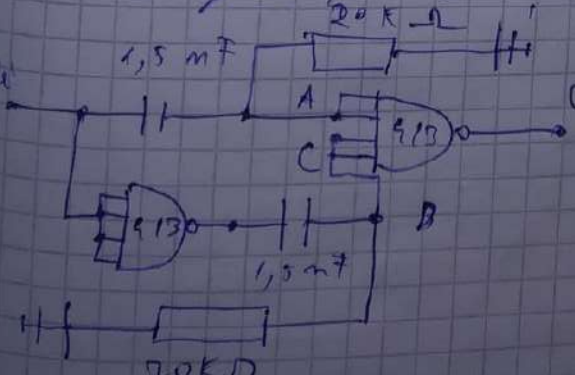



b) (*)

a) • $t = 3,5V$; durata apăsării 2 μs; durată impuls 1 μs; front ridicare coborâre 10 ms.
• osciloscopul semnal de intrare în P și de ieșire
• măsurare timpi întârziere

b) aceleași specificații și semnal.

3.9

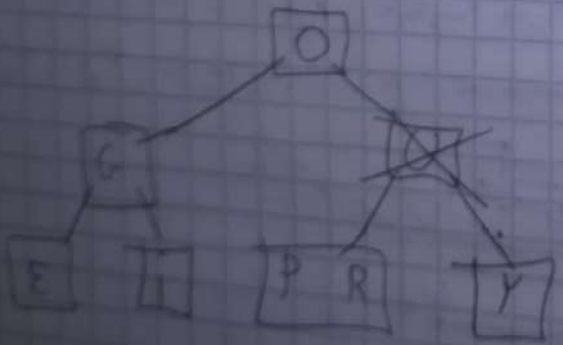


(A) (3.3) A = 3,5 V, rezoluție 10 μs, durata 5 μs, frontul 10 ns; osciloscopul de intrare în P și ieșire semnal

(3.3) semnal natangular; durată rezoluție 14 μs durată 70 μs; frontul 10 ns; osciloscopul semnal de la intrare, ieșire și din A, B, C. Valorile le sincronizăm osciloscopul

TEMA: N = 1 → V, O

1)



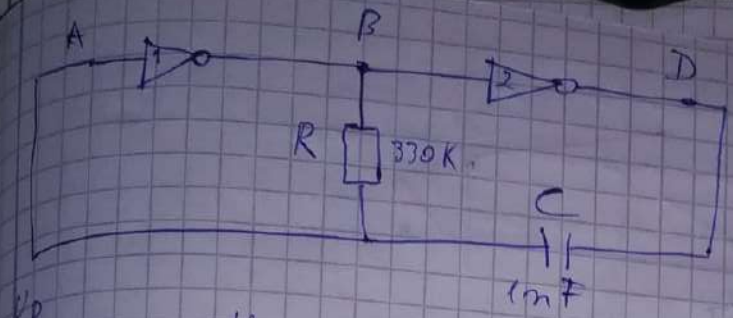
| | | |
|---|---|---|
| 3 | 26866 | 57724 |
| 4 | 26866 | 26866 |
| | 1C1200 | 0 |

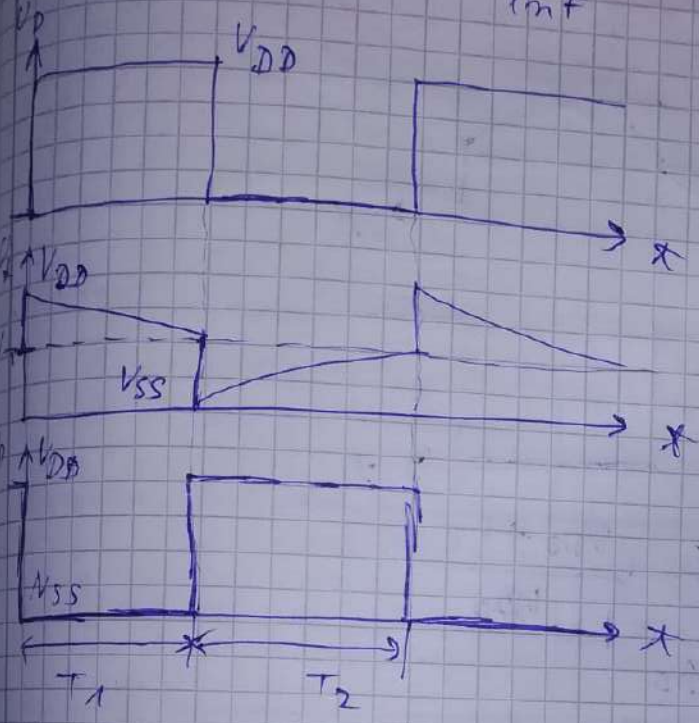
## Aplicația 7 - Circuitul:

1. Scop: studiul circuitelor basculante astabile în tehnologie CMOS (prezentare + analiza principalelor caracteristici).

2. Teoria lucrării:

2.1. CDA cu două invertoare CMOS:



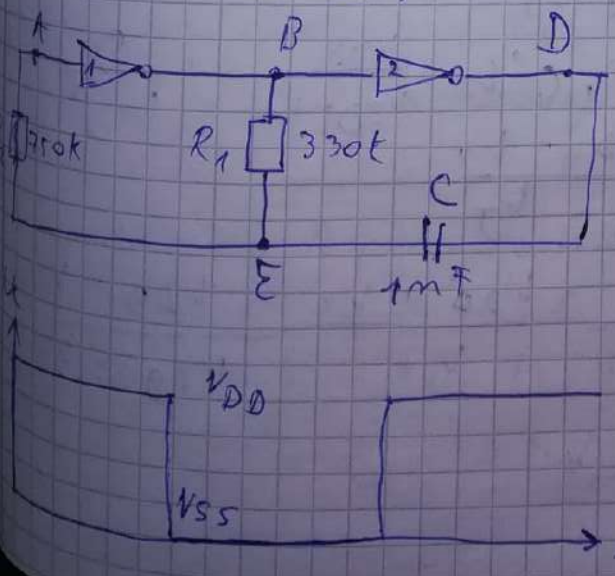



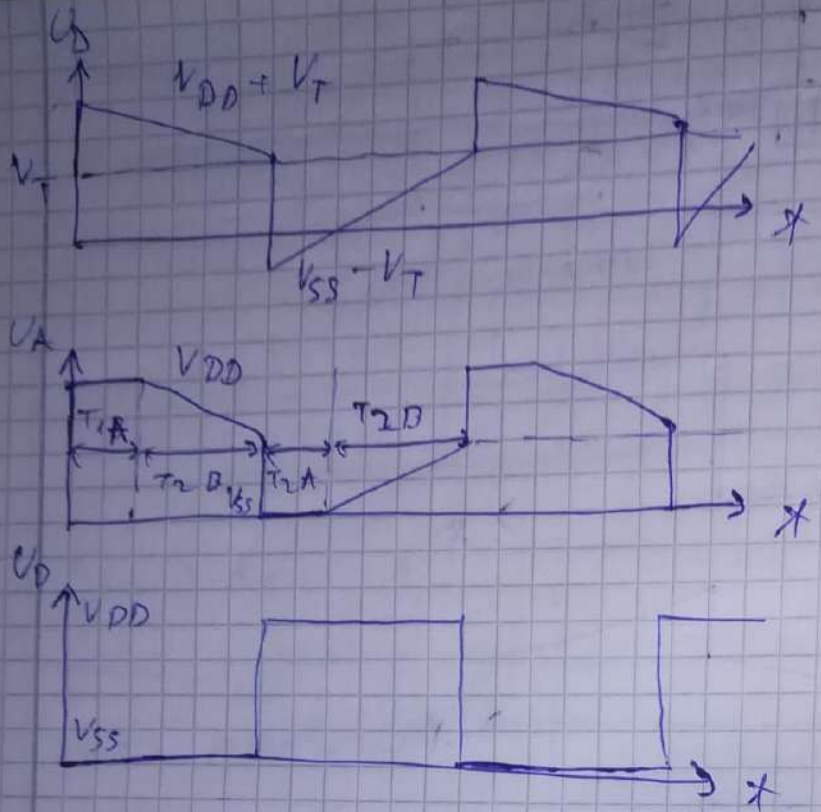

$T_1 + T_2 = T$

$$T_1 = RC \ln \frac{V_{DD}}{V_T}$$

$$T_2 = RC \ln \frac{V_{DD}}{V_{DD} - V_T}$$

2.2. CBA independent de variatiile tensiunii de alimentare.






$$T_{1A} = \frac{R_1 R_2}{R_1 + R_2} \cdot \ln \frac{V_{DD} + V_T}{V_{DD}}$$

$$T_{1B} = R_1 C \ln \frac{V_{DD}}{V_T}$$

$$T_1 = T_{1A} + T_{1B} = R_1 C \left( \ln \frac{V_{DD}}{V_T} + \frac{R_2}{R_1 + R_2} \cdot \ln \frac{V_{DD} + V_T}{V_{DD}} \right) \simeq$$

$$\simeq R_1 C \ln \frac{V_{DD} + V_T}{V_T}$$

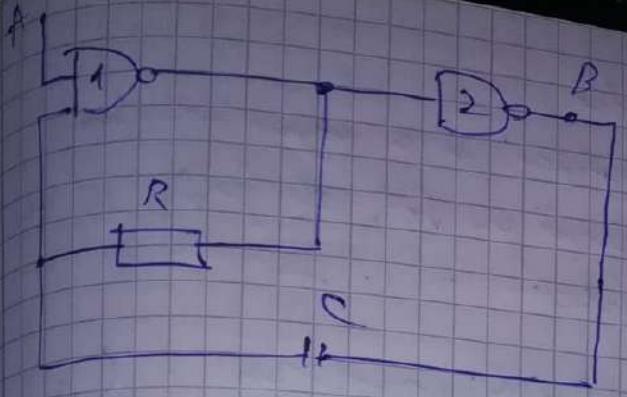
$$T_{2A} = \frac{R_1 R_2}{R_1 + R_2} \cdot C \ln \frac{2V_{DD} - V_T}{V_{DD}}$$

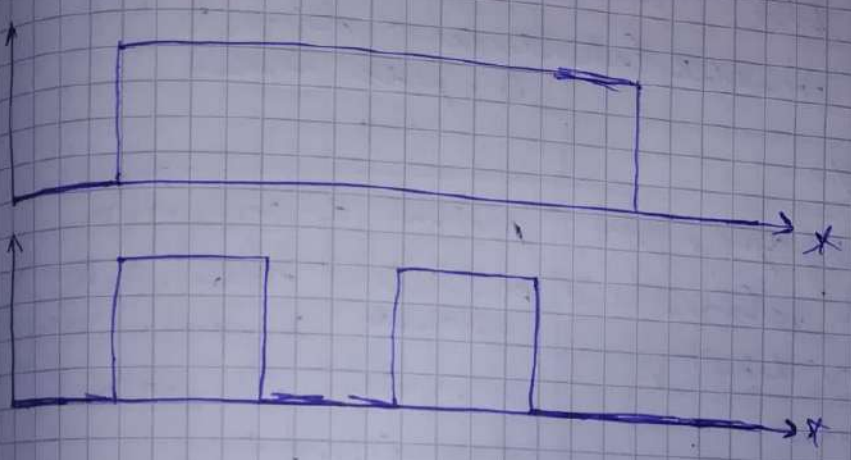
$$T_{2B} = R_1 C \ln \frac{V_{DD}}{V_{DD} - V_T}$$

$$T_2 = T_{2A} + T_{2B} = R_1 C \left( \ln \frac{V_{DD}}{V_{DD} - V_T} + \frac{R_2}{R_1 + R_2} \cdot \ln \frac{V_{DD} - V_T}{V_{DD}} \right) \simeq \ln \frac{2V_{DD} - V_T}{V_{DD} - V_T}.$$

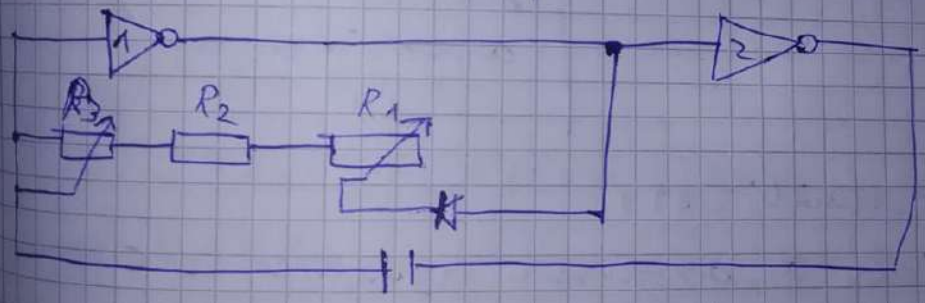
$$T = T_1 + T_2 = R_1 C \ln \frac{V_{DD} + V_T}{V_T} \cdot \frac{2V_{DD} - V_T}{V_{DD} - V_T}.$$

2.3 CBA comandat




2.4 CBA cu controlul factorului de umplere




3. Specificații & Mod de lucru:

3.1. Montaj pt. schema 1 (în loc de inversoare → porți logice) $V_{DD} = 5V$; $R = 330k\Omega$, $C = 1nF$ - $T_1, T_2, T_3 = ?$ + osilografiere.

3.2 Montaj schema 2, $R_1 = 320k\Omega$, $C = 1nF$, $V_{DD} = 5V$, $R_2 = 750k\Omega$. $T_1, T_2, T = ?$ + osilografiere semnal.

3.3 $V_{DD} = 10V$ $T_1, T_2, T = ?$; comparare rezultate pt. cele pt. $V_{DD} = 5V$.

Mod de lucru:

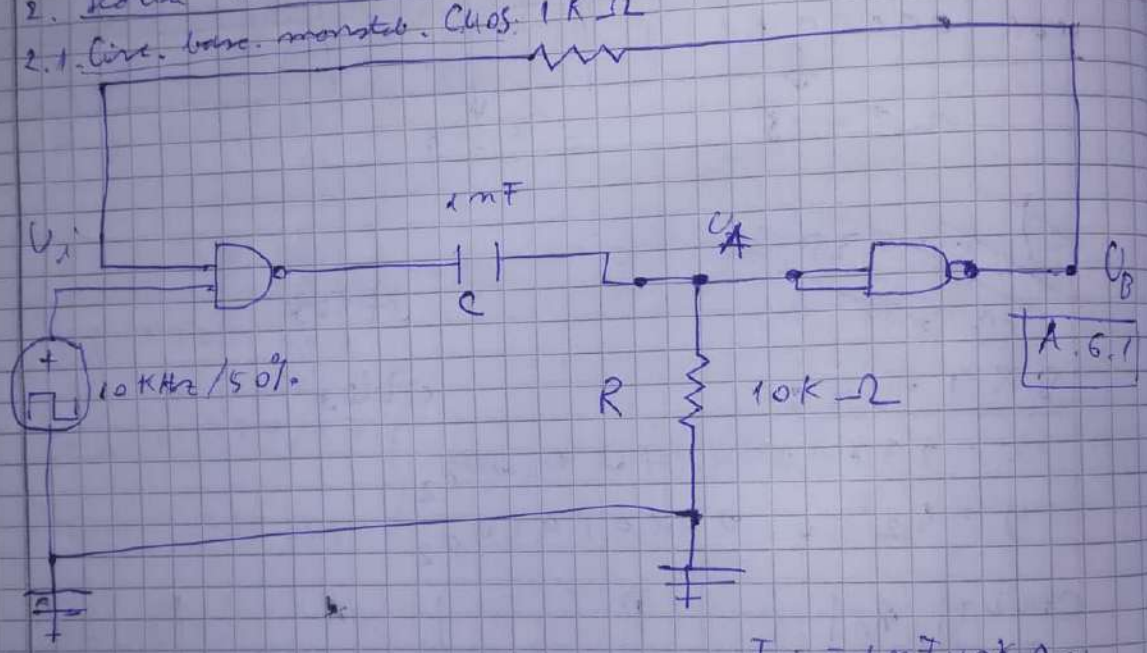
| SHIFT | K=2 / 000001011 | 000001011 |
|---|---|---|
| | Rest = 7 7 | Cat = 5 |

Aplicația 8 (i) (compus):

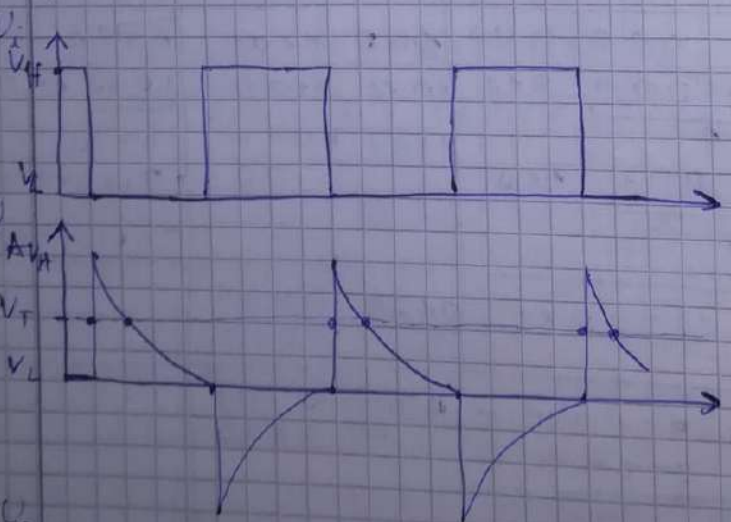
1. Scop: studiul circuitelor basculante monostabile realizate cu porți de tip ȘI-NU (AND).

2. Teoria lucrării:

2.1. Circ. basc. monostab. CMOS. 1 KΩ




A.6.1




$T_1 = 1nF \cdot 10K\Omega \cdot \ln \frac{5V}{1,5V} = 10^{-9} \cdot 10 \cdot 10^{3} \cdot \ln \frac{5}{1,5} =$

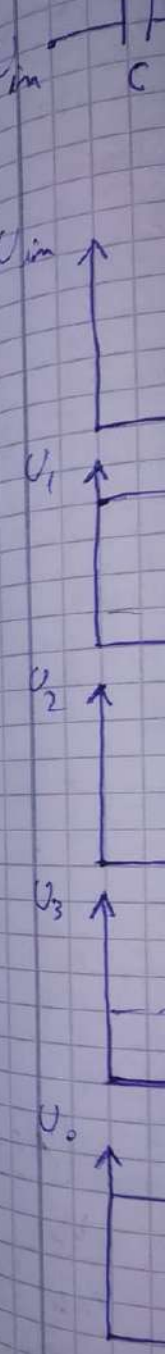
$= 1,2 \cdot 10^{-5} s$

$= 12 \mu s.$

$T_1 = RC \ln \frac{V_{DD}}{V_T}$

A.G.2

2.2.

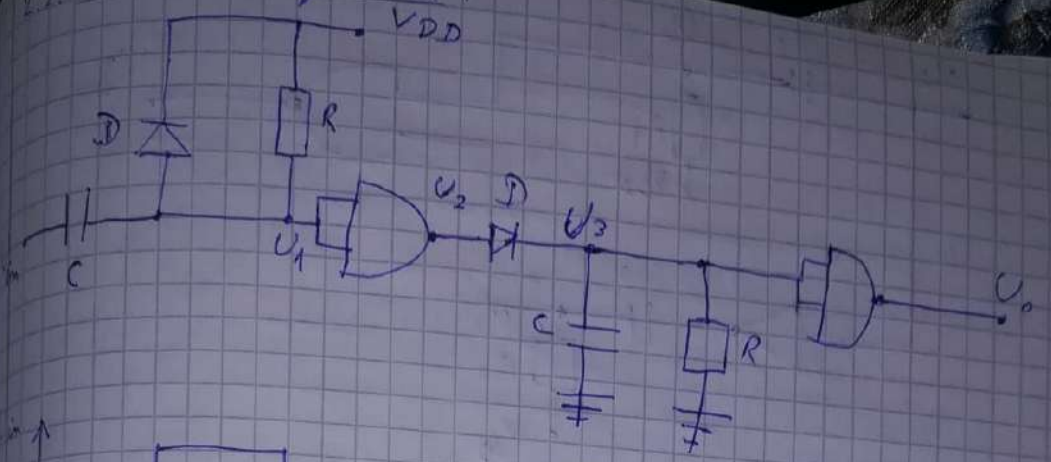


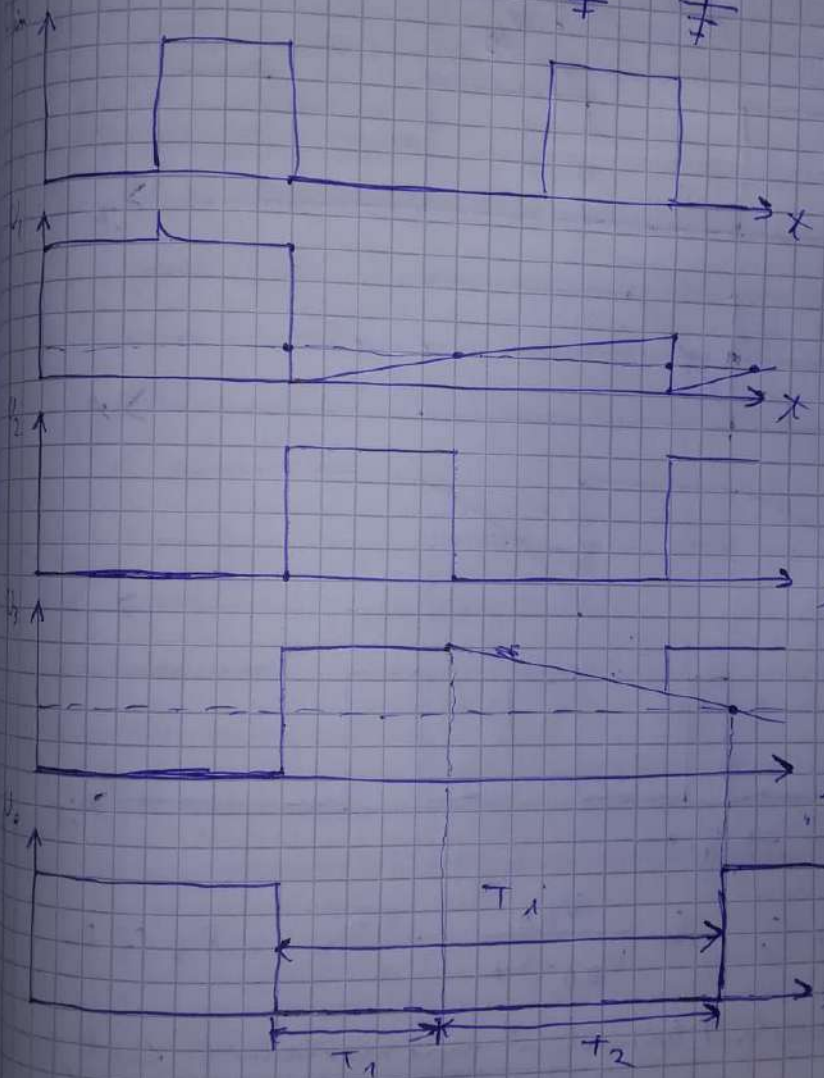

$T_1$

$T_2$

2.3

2.2. C.B.M. comunicat:



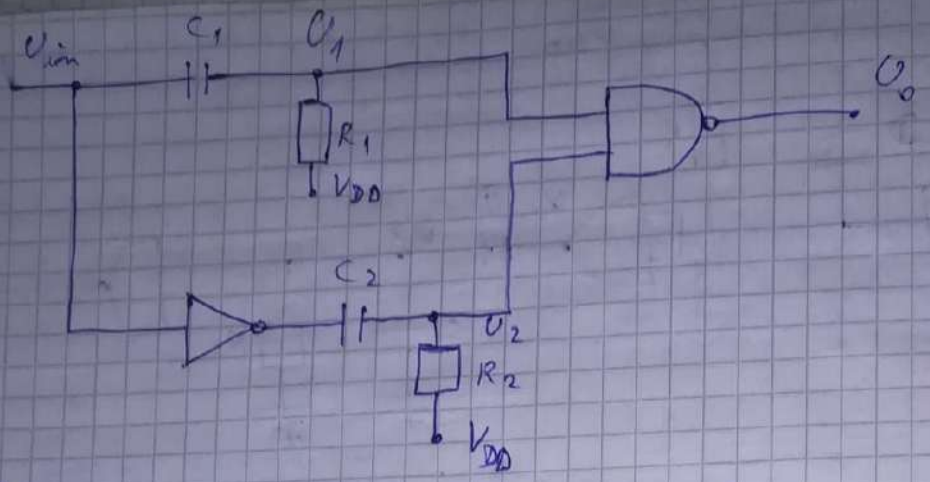



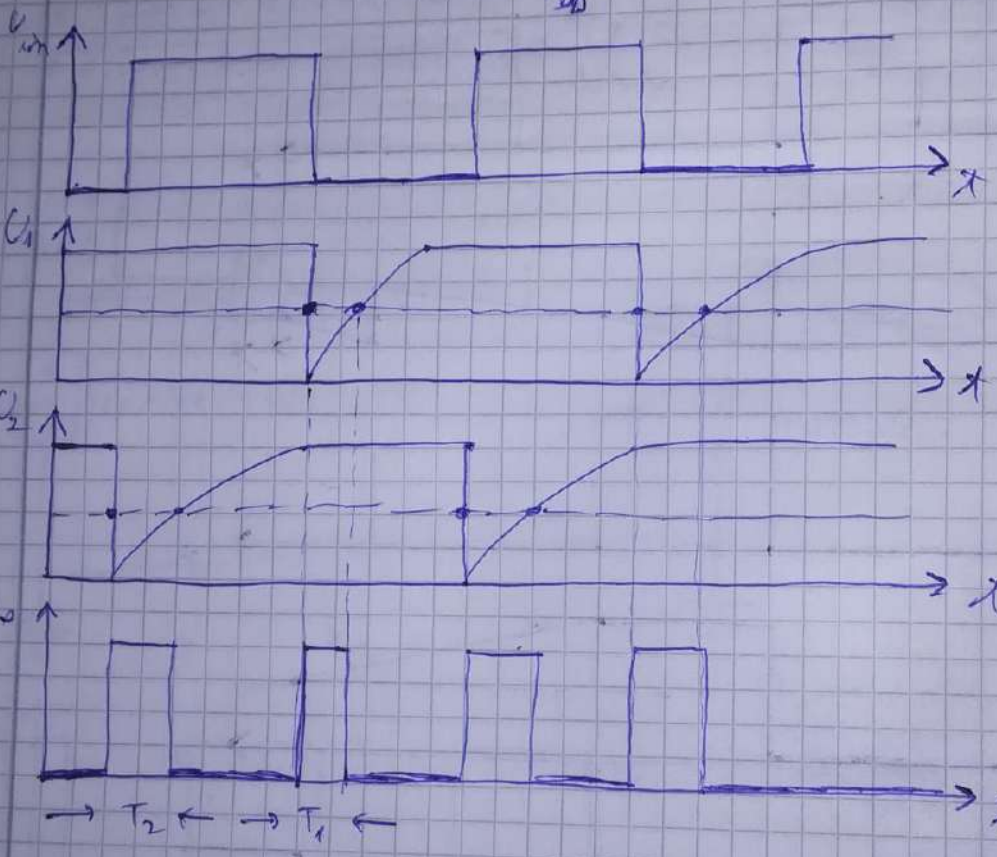

A.6.3

$T_1 = R_1 C_1 \ln \frac{V_{DD}}{V_{DD} - V_T}$ $\quad T_i = T_1 + T_2$

$T_2 = R_2 C_2 \ln \frac{V_{DD}}{V_T}$

Dublor de frecvență:

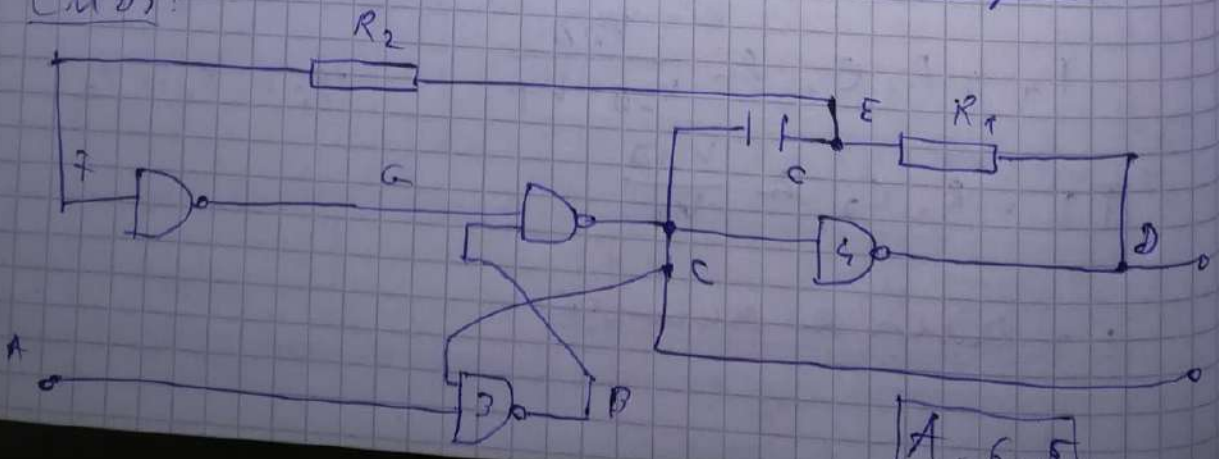


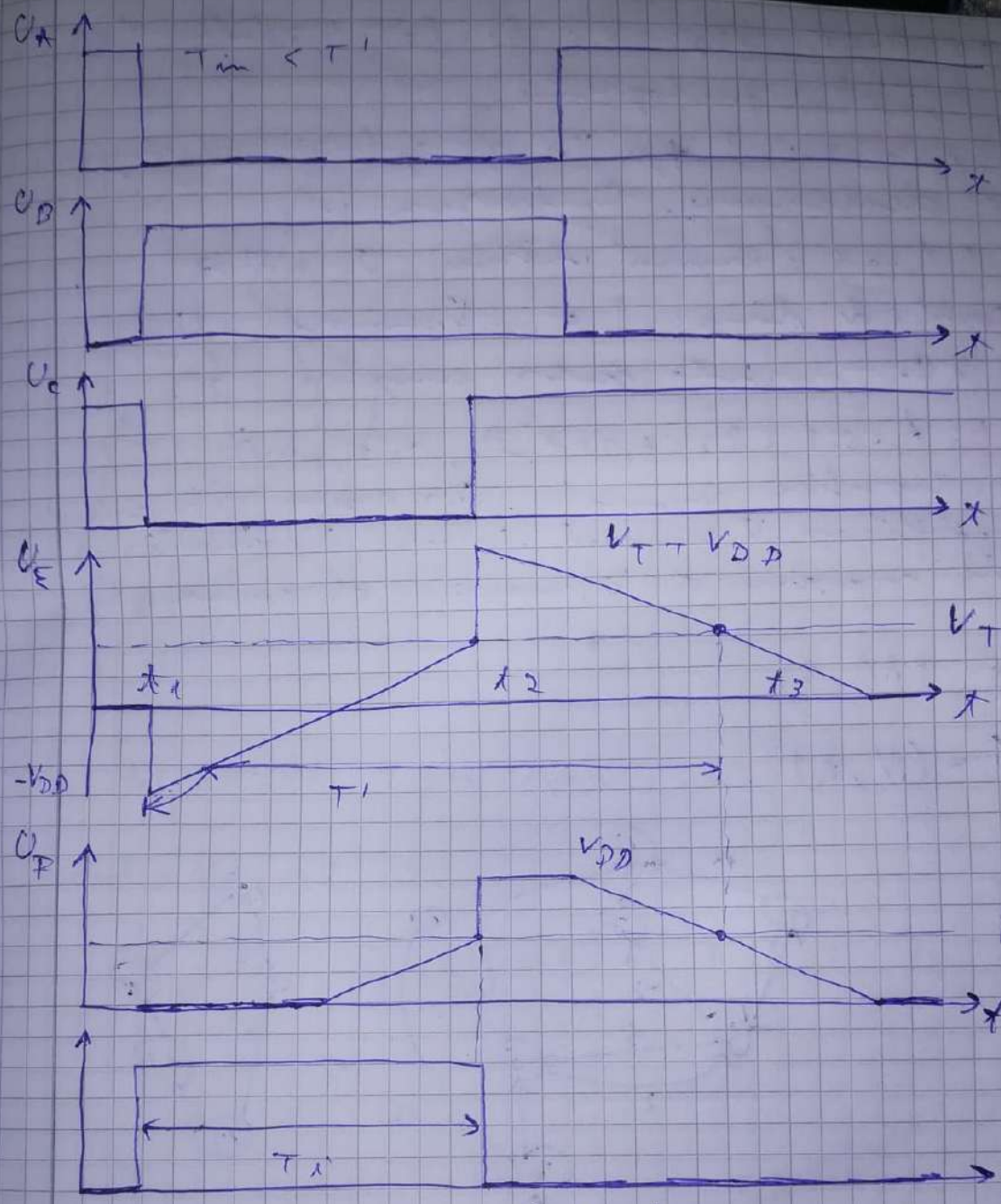

A.6.4

$T_1 = R_1 C_1 \ln \frac{V_{DD}}{V_{DD} - V_T}$ ; $T_2 = R_2 C_2 \ln \frac{V_{DD}}{V_{DD} - V_T}$

2.4. Circuitul basculant monostabil cu 4 porți CMOS:




A.6.5




A.6.4

$$T_i = R_1 C \ln \frac{V_{DD} - (-V_{DD})}{V_{DD} - V_T}$$

A. 6. 6

$t_2 - t_1 = T_i$

$t_3 - t_2 = R_1 C \ln \frac{V_{DD} - V_T}{V_T} = T_2$

cond: $T_i \leq T_{im} \leq T_i + T_2$

3. Mod de lucru:

3.1 A.6.1 ; $V_{DD} = 5\,V$ ; $R = 4{,}7\,k\Omega$ ; $C = 2{,}2\,nF$.

Osilograme, $T_i \leq$ semnal spuitie.

$T_i = ?$ comparam cu val teoretic.

3.2 A.6.3.; $V_{DD} = 5V$; $R_1 = R_2 = 4,75 k\Omega$; $C_1 = C_2 = 2,2 nF$; oscilograme $T_1, T_2, T_{ac} = ?$ și $T_{ac}$ = ciclul unui dinte de declanșare. Reținem $T_{ac}$

3.3. A.6.4; semnal intrare $V = 1 MHz$ const. de timp = ? a.i. $V_{semnal\ ieșire} = 2 MHz$, prin ce factor umplut = $\frac{1}{2}$.

3.4. A.6.5; $R_1 = 4,75 k\Omega$, $C = 2,2 nF$. oscilografia semnale. Obs. regim dinamic, modificăm V semnal intrare pt. stabilirea ratei maxime de V maxime de comandă.

Lab PAA 9:

$G = (N, A)$

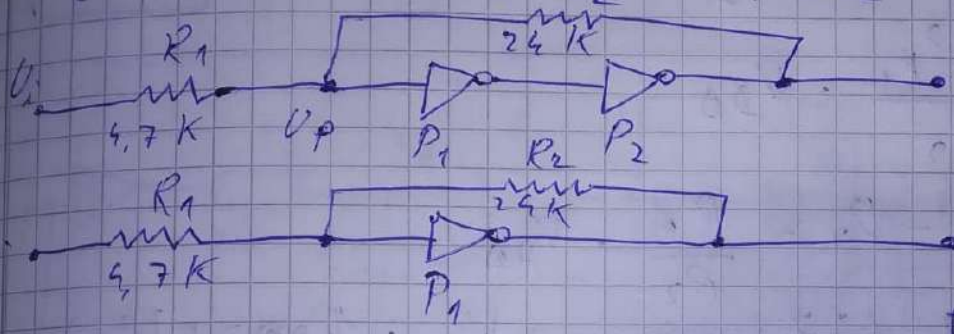
Aplicația 9 C1 (conținut):

1. Scop: studiul triggerelor Schmitt cu porți CMOS.

2. Teoria lucrării:

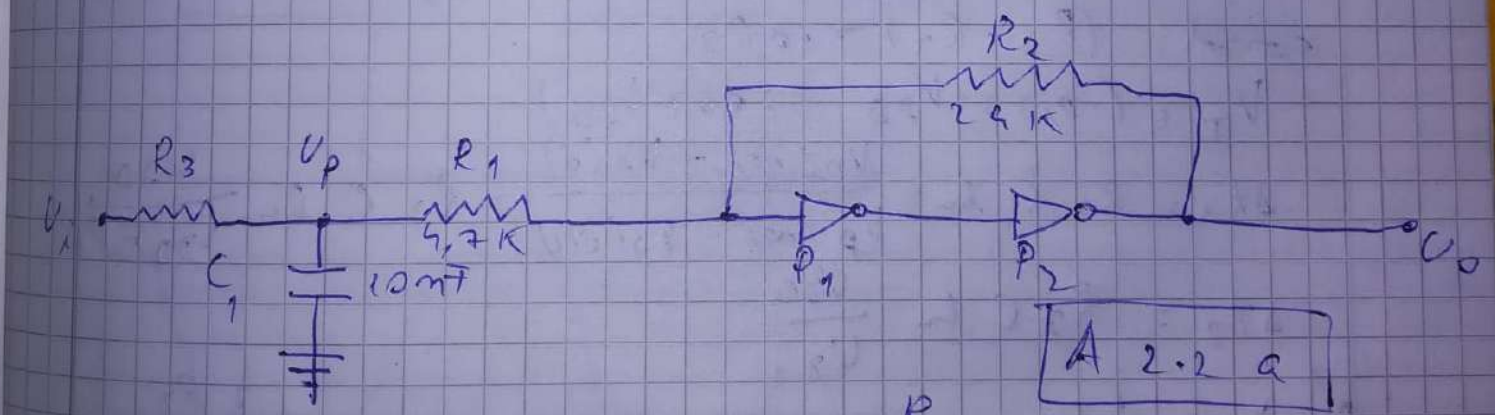
$$U_1 = V_T \cdot \frac{R_1 + R_2}{R_2} - V_L \cdot \frac{R_1}{R_2}$$

$$U_2 = V_T \cdot \frac{R_1 + R_2}{R_2} - V_H \cdot \frac{R_1}{R_2}$$

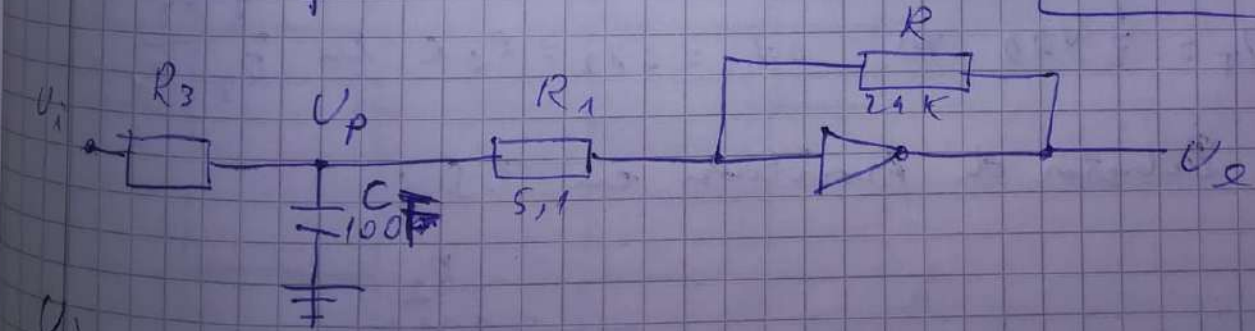


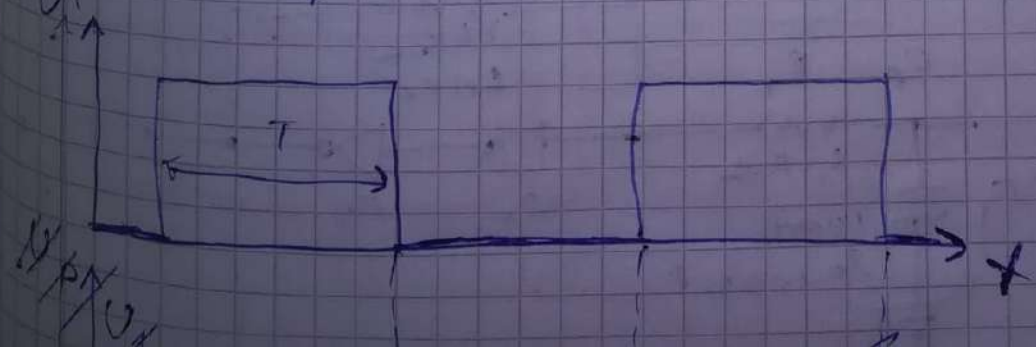

A 2.1

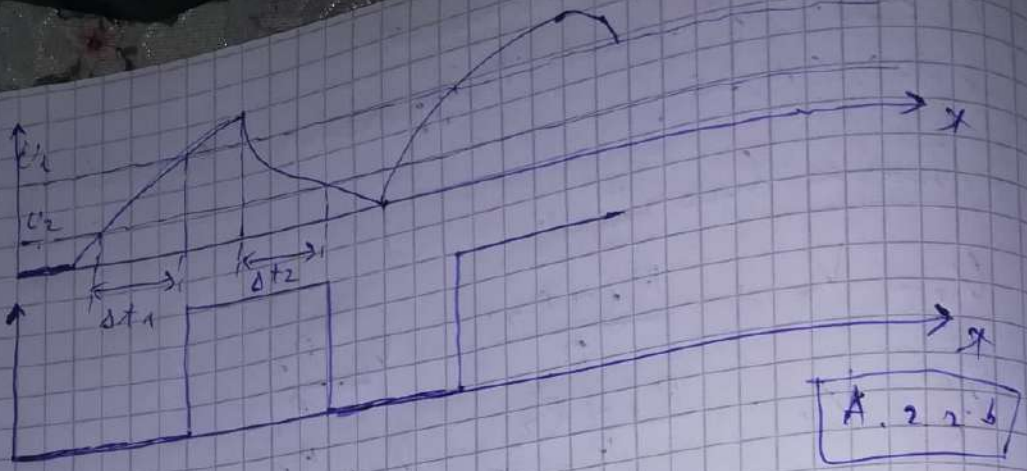
Pt. $V_L = 0V$; $V_H = V_{DD}$; $V_T = \frac{1}{2} V_{DD}$ ⇒




A 2.2 a






A. 2. 2. b

$$U_1 = \frac{R_1 + R_2}{2R_2} \cdot V_{DD}$$

$$U_2 = \frac{R_2 - R_1}{2R_2} \cdot V_{DD}$$

$$U_H = U_1 - U_2 = \frac{R_1}{R_2} \cdot V_{DD}$$

tensiune de histerezis $= 1\,V$ ; $V_{DD} = 5\,V$

2.1. Circuit cu întârziere cu trigger Schmitt

cond: $(R_1 + R_2) \geq 10 R_3$

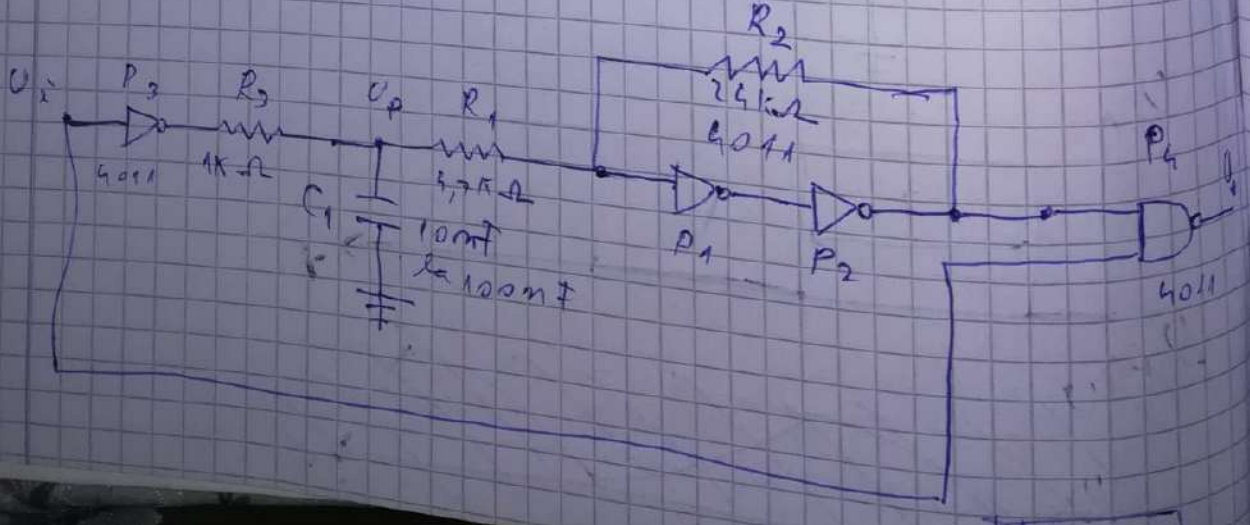
$V_T \in (0{,}33\, V_{DD} \; ; 0{,}67\, V_{DD})$

$$\Delta t_1 = R_3 C \ln \frac{V_p(\infty) - V_p(0)}{V_p(\infty) - V_p(\Delta t_1)} = R_3 C \ln \frac{V_{DD}}{V_{DD} - V_1}$$

$$\Delta t_2 = R_3 C \ln \frac{V_{DD}}{U_{21}}$$

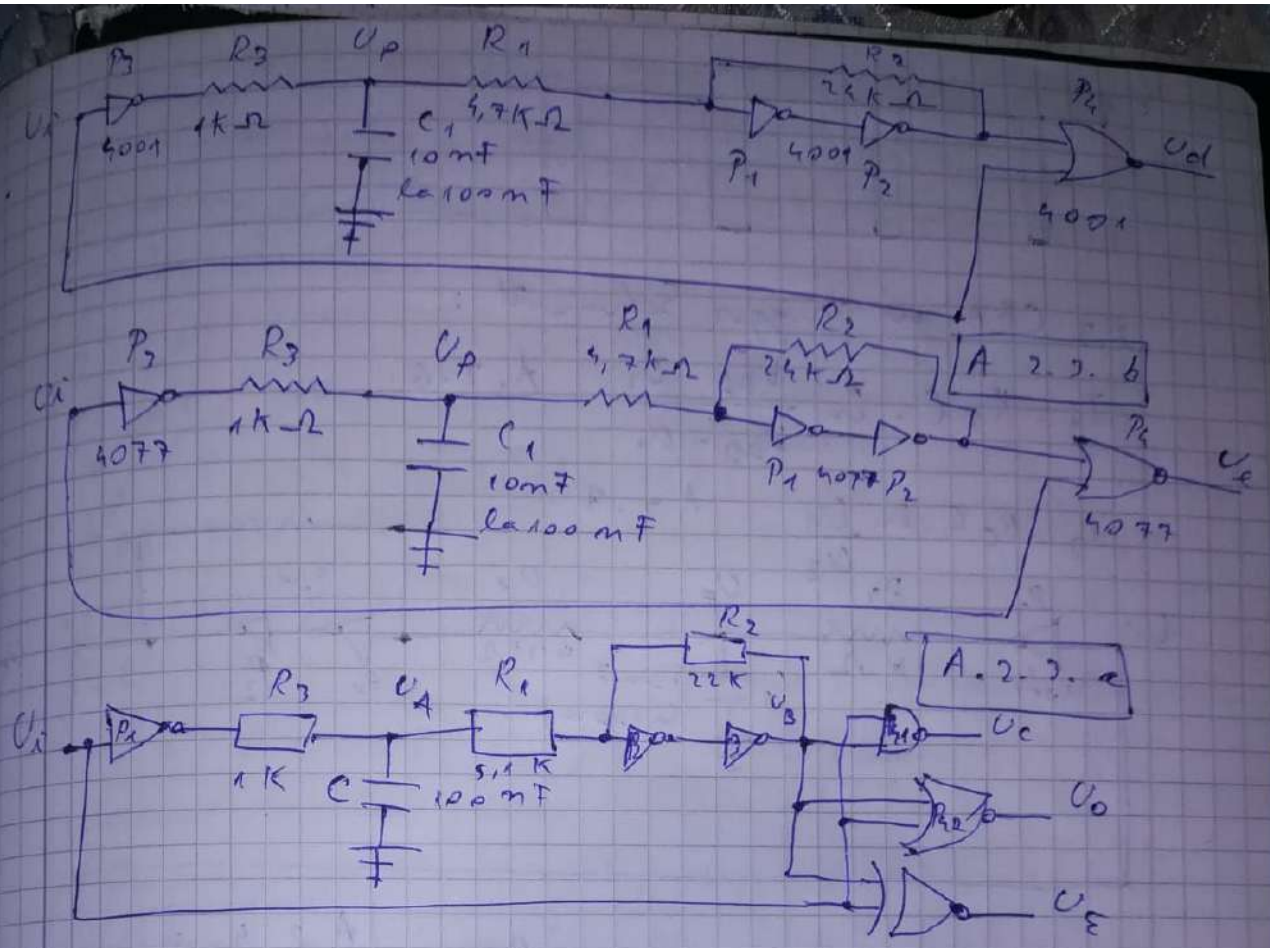
Pt. $V_T = \frac{1}{2} V_{DD} \Rightarrow \Delta t_1 = \Delta t_2 = R_c C \ln \frac{2R_2}{R_2 - R_1}$

2.2. Detector de frontură cu T.S. :

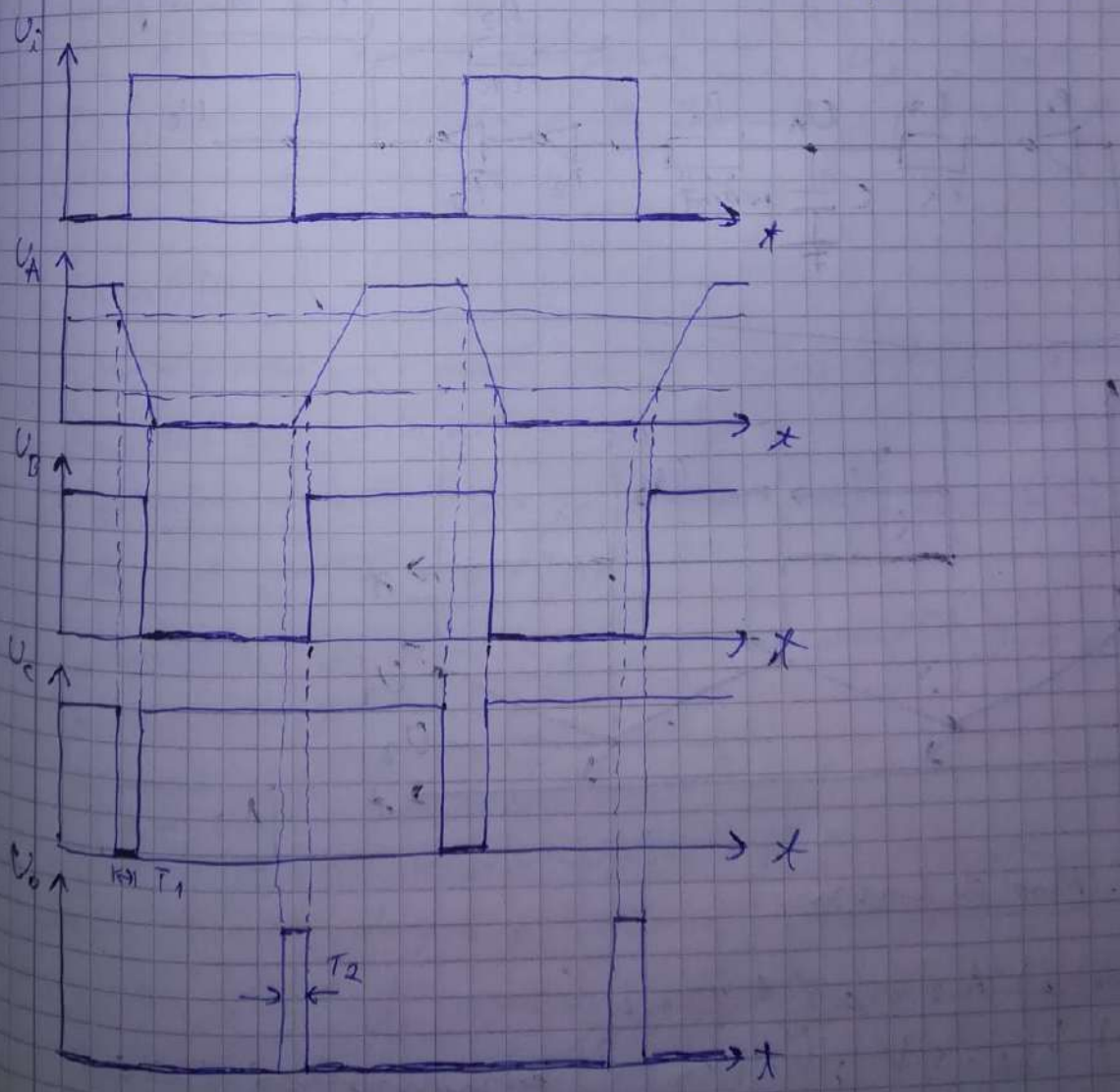



A 2. 3. a




$P_3$
$R_3$
1 kΩ
4001
$U_p$
$R_1$
4,7 kΩ
$C_1$
10 nF
la 100 nF
$R_2$
22 kΩ
$P_1$
$P_2$
$P_4$
$U_{el}$
A 2.3.b
4077
$U_e$


$R_2$
22 K
A.2.3.c
$R_3$
1 K
$U_A$
$R_1$
5,1 K
100 nF
$U_B$
$U_C$
$U_D$
$U_E$

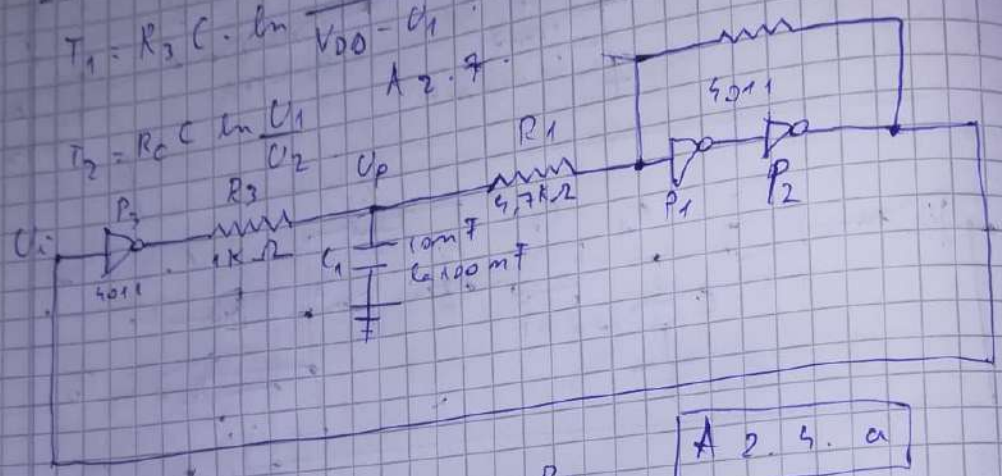

$U_i$
$U_A$
$U_B$
$U_C$
$U_D$
$T_1$
$T_2$
t



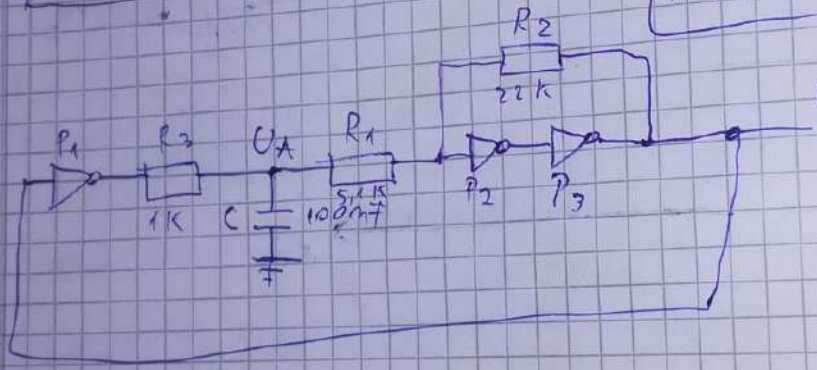

2.3 CBA cu trigger Schmitt:

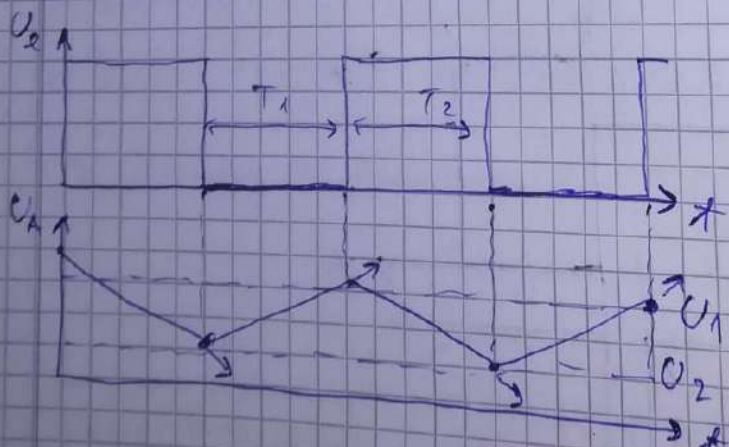
$T_1 = R_3 C \cdot \ln \frac{V_{DD} - U_2}{V_{DD} - U_1}$ A.2.6.

$T_2 = R_2 C \ln \frac{U_1}{U_2}$ A.2.7.




A 2.4. a




A 2.4 b

3. Mod de lucru:

3.1. a) A 2.1 a / A 2.1 b

b) $V_{DD} = 5V$; $U_1$; $U_2$ = ? caracteristică statică

$U_1 \in [0V; 5V]$, increment $+0,1V$.

- umărirea $U_2$, const. la 0V.

funcția prog $U_2$, $U_2 \in [5V; 0V]$, increment $-0,1V$

c) $V_T$ = ?

3.2. a) A.2.2

b) $\Delta t_1$, $\Delta t_2$

$T_i = T/2 = 2 \max(\Delta t_1, \Delta t_2)$

$V_{DD} = 5V$

c) osciloscopiem $U_i$, $U_P$, $U_c$

Calc. $T_i$, $T$, $\Delta t_1$, $\Delta t_2$. Comparam cu cele teoretice.

3.3. a) A.2.3 ($P_1$, $P_2$ pct. anterioare / semnal rectangular

b) Osciloscopiem $U_i$, $U_A$, $U_B$, $U_C$

Obținem cea și $U_D$ $U_E$. $U_c$, $U_i$; Calc. $T_1$, $T_2$; comparam cu date teoretice (A.2.6, A.2.7)

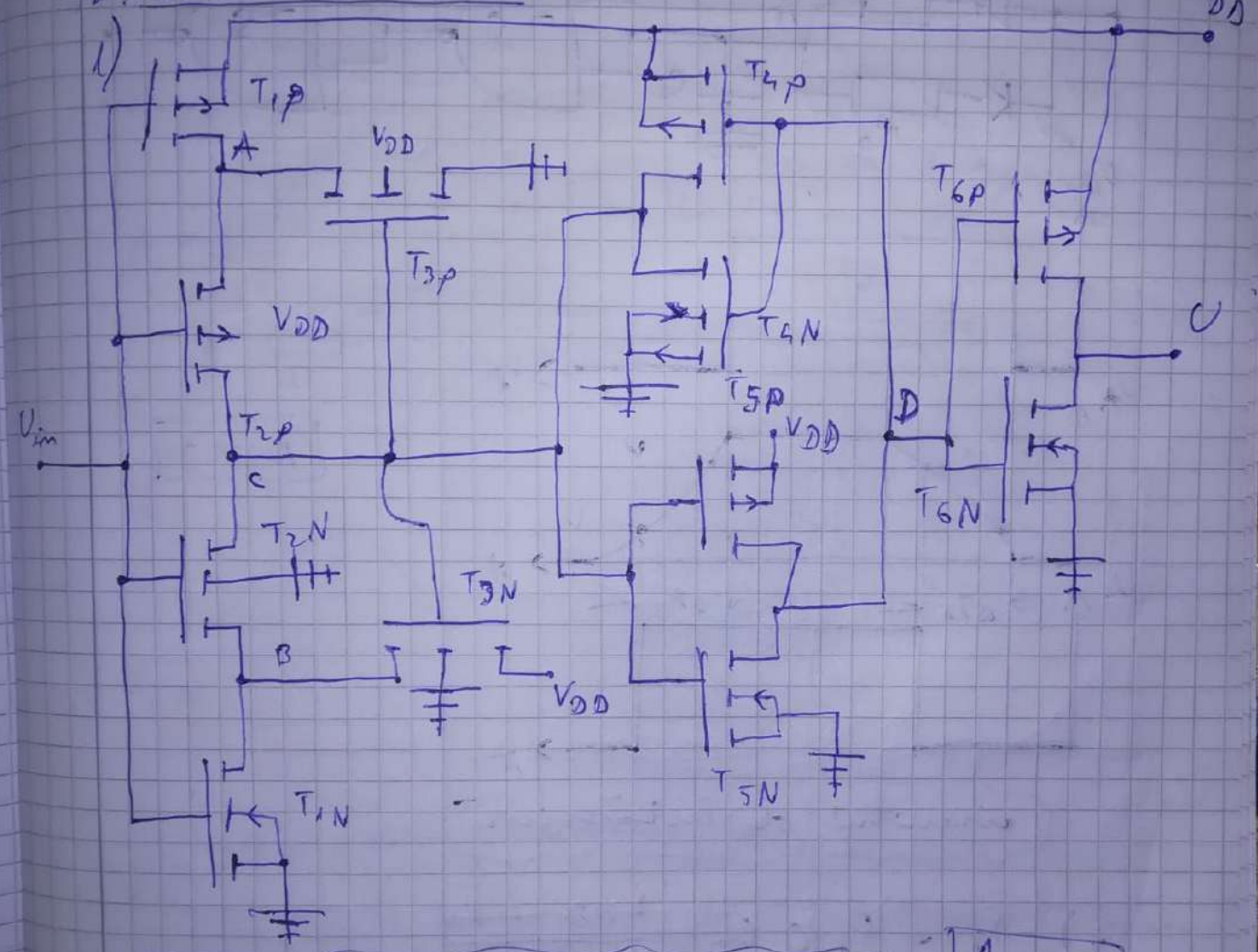
$U_1$, $U_2$ - date la 3.1.

Curs BD 10:

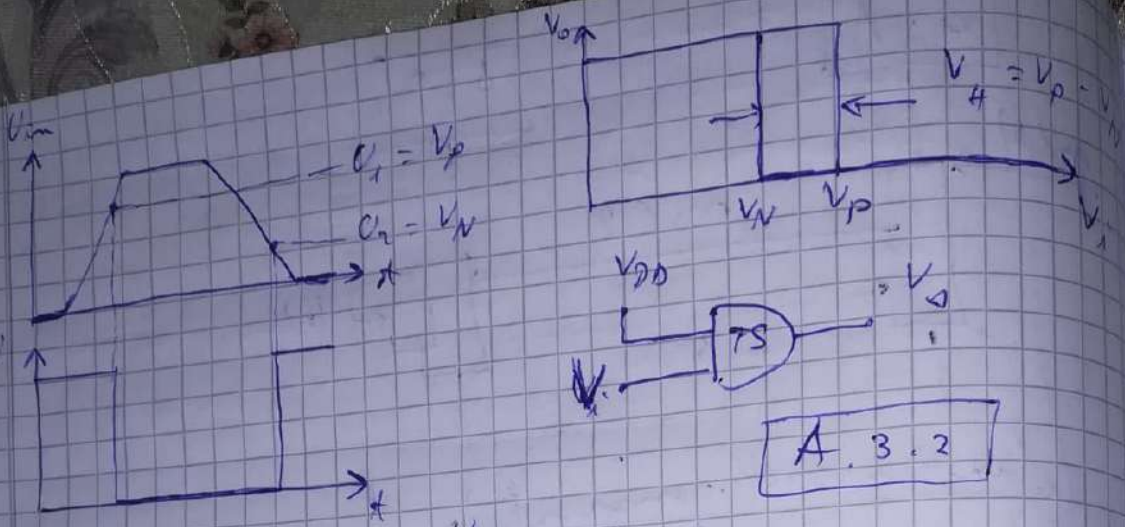
Aplicația 11 (complet):

1. Scop: studiul triggerelor ~~int~~ Schmitt integrați CMOS.

2. Teoria lucrării:



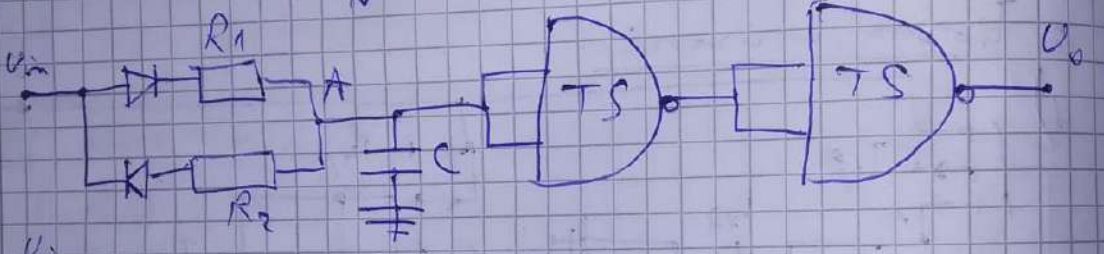

{trigger Schmitt integrat CMOS} A. 3. 1

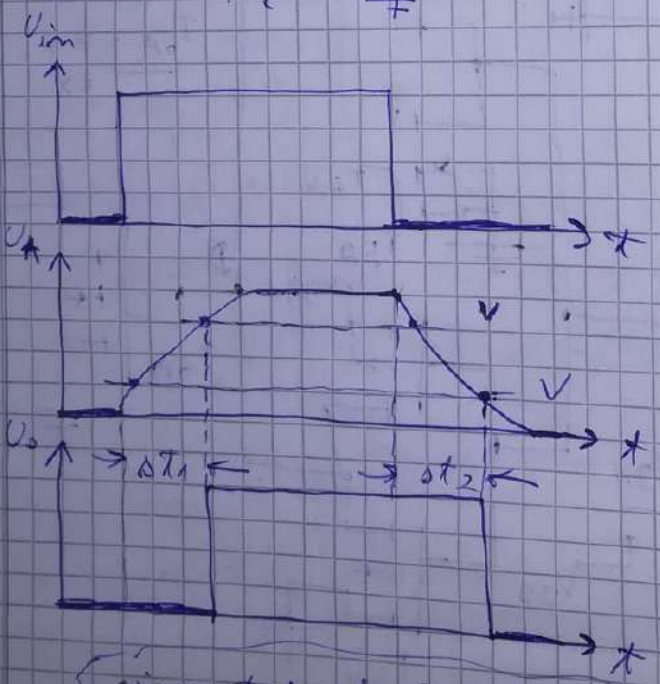



A.3.2

2) $\Delta t_1 = R_1 C \ln \frac{V_{DD}}{V_{DD} - V_P}$ } intervine

$\Delta t_2 = R_2 C \ln \frac{V_{DD}}{V_N}$



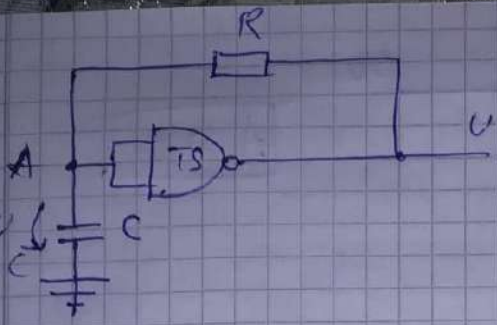

A 3.3

circuitul de întârziere

3) $T_1 = R_1 C \ln \frac{V_{DD} - V_N}{V_{DD} - V_P}$

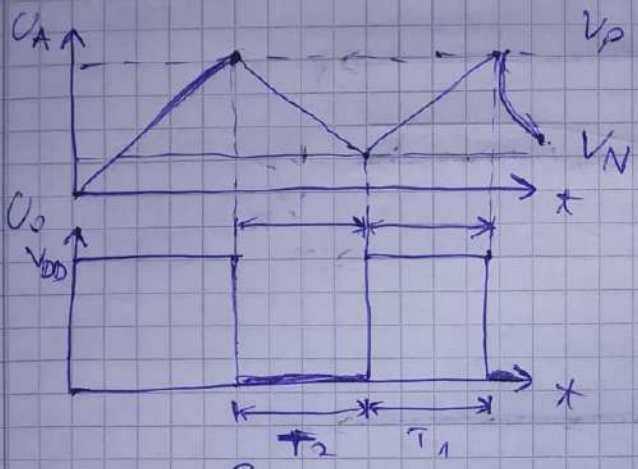
$T_2 = R_2 C \ln \frac{V_N}{V_P}$

unde $T = T_1 + T_2$ - semnal repetitiv

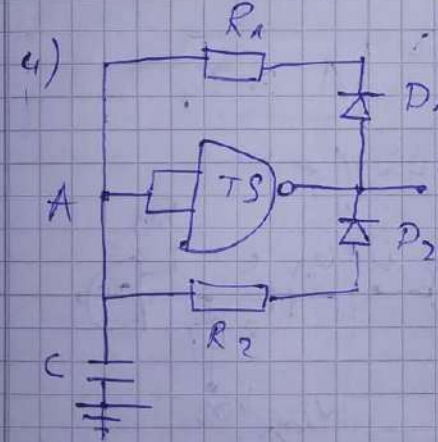



CDA cu trigger Schmitt

A.3.4

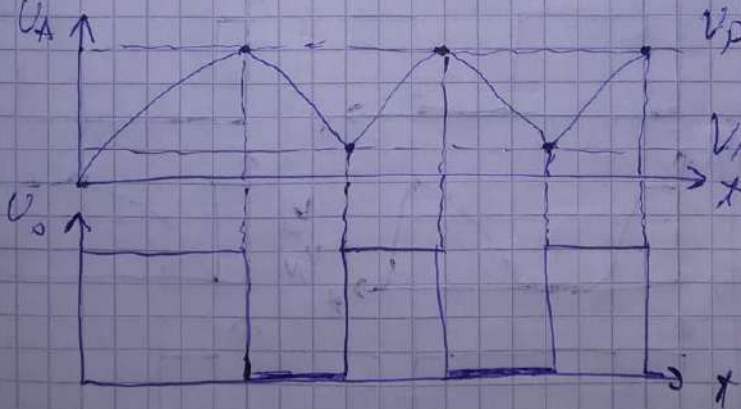



4)




CBM cu trigger Schmitt

A.3.5

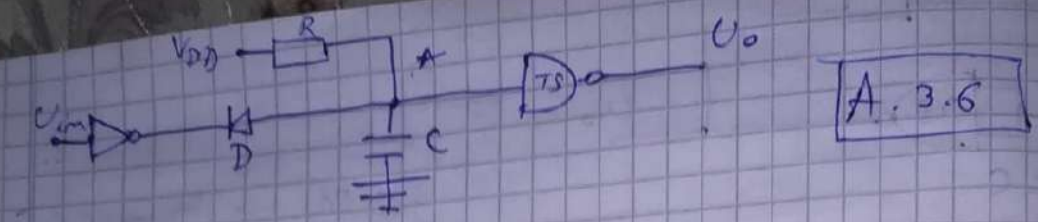



$T_i$ - impuls intrare $\Rightarrow T_i + T_1$ - impuls ieșire

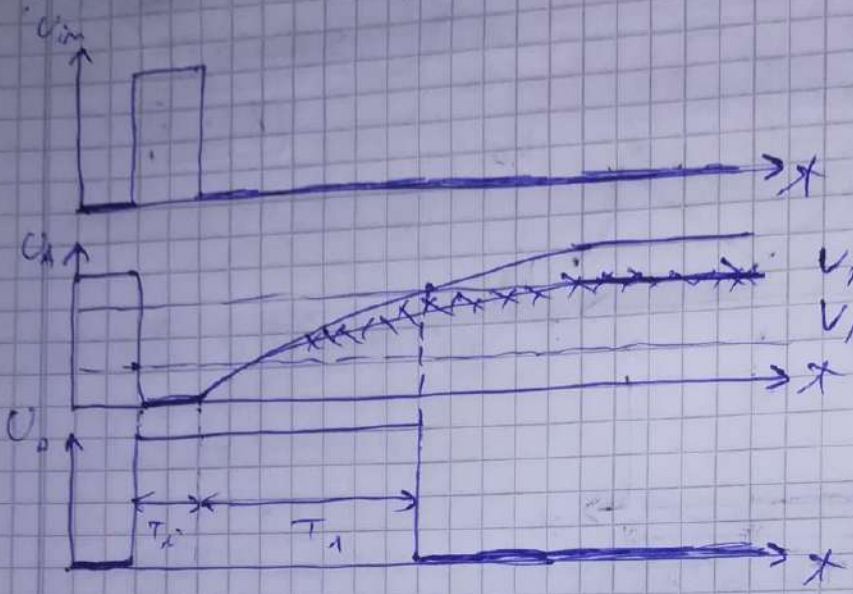
unde $T_1 = RC \ln \frac{V_{DD} - V_D}{V_{DD} - V_P}$

$V_D = 0{,}75\,V$ (căderea de tensiune directă)

$I_{max} = \frac{C V_{DD}}{T_i} + \frac{V_{DD}}{R}$

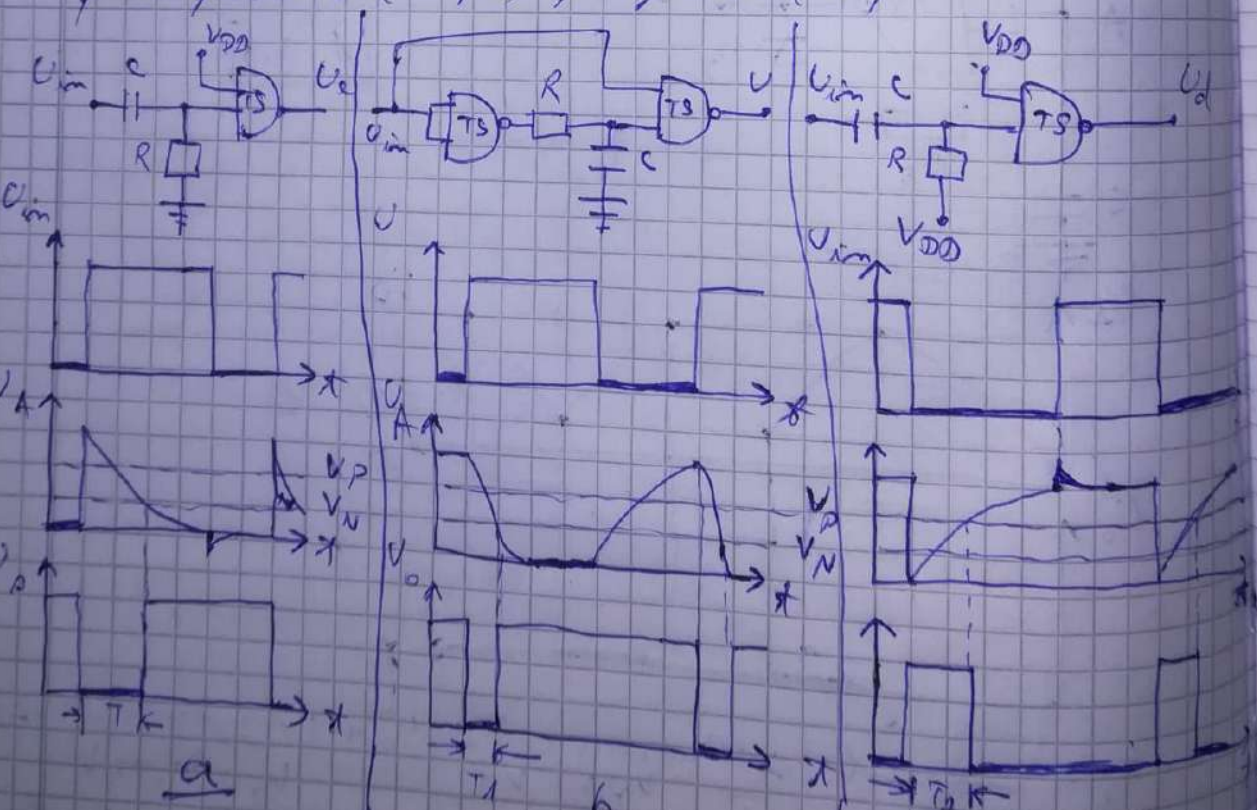



A. 3.6

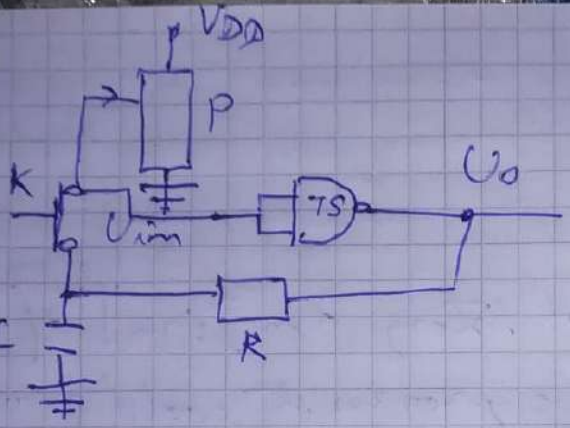



5) detector de front — A. 3.7

front pozitiv (a, b); negativ (c)



6)




Comutator cu trigger Schmitt

A.3.8

## 3. Mod de lucru:

3.1 caracteristica experimentală $U_0 = f(U_i)$ a trăgătorului Schmitt ($V_{DD} = 5V$). tensiuni de prag = ?

3.2 A.3.3; $\Delta T_1, \Delta T_2$ = ?; osciloscopierea semnalelor intrare, ieșire; $R_1 = 10k\Omega$, $R_2 = 24k\Omega$, $C = 2,2nF$, $V_{DD} = 5V$; $T_1 \Rightarrow \Delta T_1$ și $T - T_1 \Rightarrow \Delta T_2$

3.3. $R = 10k\Omega$, $C = 100pF$, $V_{DD} = 5V$; A.3.4; osciloscopierea și determinarea $T_1$, $T_2$

3.4. A.3.5; $R = 10k\Omega$; $C = 100pF$, $V_{DD} = 5V$; — " — ; $T - T_1 \Rightarrow T_1$.

3.5. A.3.6; $C = 100pF$; $R_1, R_2$ dimensionați a.î. $T_1 = T_2 = 3\mu s$. — " — (de la 3.4).

3.6. A.3.7; $R = 1k\Omega$, $C = 100pF$; osciloscopierea și determinarea timpi. conclusii (diferențe și asemănări între trigger Schmitt din familia TTL și (MOS).